

THINKING IN CHESS

An Intuitive Game-Theoretic Textbook

How to model positions, read incentives, search trees, manage risk, and learn from opponents

A chess move is not an isolated object. It is an action inside a living tree of replies. This book teaches you to see that tree without drowning in it.

Original educational synthesis prepared from the game-theory framework supplied by the reader

Thinking in Chess

Edition 1.0

This is an original textbook. It does not reproduce the wording, examples, exercises, or chapter text of the uploaded game-theory book. Instead, it translates a broad sequence of ideas—game frames, preferences and utility, strategic choice, extensive-form reasoning, uncertainty, beliefs, and incomplete information—into a practical language for chess.

All chess diagrams and examples in this book are pedagogical constructions unless otherwise stated. Piece values and probability estimates are heuristics, not laws. In any legal dispute over a position, the official rules of the event govern.

Preface: The move is not the point

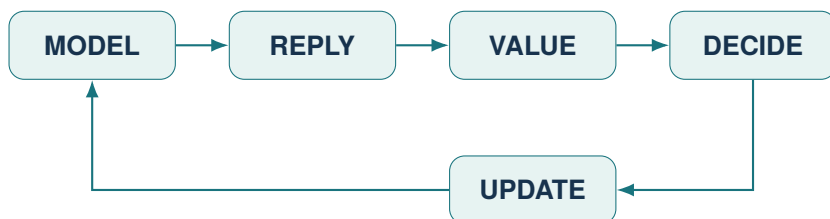
Most chess instruction presents answers: a tactic, a rule, an opening move, a famous plan. Answers are useful, but they decay when the position changes. A way of thinking travels farther.

Game theory begins from a deceptively simple observation: your choice cannot be evaluated in isolation from someone else's choice. Chess makes that observation visible. Every move changes the actions available to your opponent, the value of future positions, and the information both players reveal about themselves. The board is fully visible, yet the decision remains uncertain because neither player can calculate everything and neither knows exactly what the other has noticed.

This book therefore treats chess as a sequence of structured decisions. You will learn to ask five questions:

1. **What is the state?** Record material, safety, activity, structure, time, and whose move it is.
2. **What am I trying to maximize?** The answer is not always “material.” It may be winning chances, drawing security, simplicity, time, or tournament score.
3. **What can both players do?** Generate a small set of candidates, then search for the opponent's strongest reply.
4. **How do the branches compare?** Use worst-case reasoning for forcing lines and expected-value reasoning for practical uncertainty.
5. **What did the last move teach me?** Update your model of the position and of the opponent.

The result is a compact loop:



You do not need advanced mathematics. Formal boxes make the logic precise, but every formal idea is paired with an intuition, a chess example, and a memory hook.

The aim is not to turn you into an engine. It is to help you allocate human attention well.

Who this book is for

The core reader knows how the pieces move and has encountered basic game-theory words such as strategy, payoff, and Nash equilibrium. A rules refresher is included. Beginners can read straight through; experienced players can use the chapters on minimax, uncertainty, opponent modeling, time management, and review as a separate course.

How to use it

Read one chapter at a time. Before checking a worked example, cover the continuation and name the opponent's best reply. Write answers to the exercises. During your next games, choose only one chapter habit to practice. A habit applied at the board is worth more than ten ideas recognized on the page.

The appendices contain a notation guide, printable decision sheets, complete exercise solutions, and a glossary. Chapter 26 turns the book into a twelve-week study plan.

The chapter in one minute

Chess improvement is not mainly the accumulation of more moves. It is the improvement of the questions you ask before choosing one.

A map from game theory to chess

Game-theory lens	Chess translation	Practical question
Game frame	Board, side to move, legal moves, clocks, rules	What facts define the decision?
Preferences and utility	Win/draw/loss plus practical goals	What result and what kind of position do I need?
Strategic form	Complete plans and opening repertoires	What will I do against each broad response?
Extensive form	The move tree	What reply follows each candidate?
Best response	Strongest counterplay	What would I hate to see after my move?
Dominance	One candidate is no worse in relevant branches	Can I safely eliminate a move?
Nash equilibrium	Mutually optimal strategies	What survives best play from both sides?
Backward induction	Solve from the leaf positions backward	Which forcing ending settles the line?
Expected utility	Value weighted by practical likelihood	Which move scores best against a fallible human?
Mixed strategy	Varying preparation across repeated games	Am I becoming predictable?
Beliefs and Bayesian updating	Model of opponent's style and knowledge	What did the last move reveal?
Incomplete information	Hidden preparation, intentions, fatigue	Which unknowns should affect my choice?
Repeated games	Matches, tournaments, reputation	How does today's decision change tomorrow's incentives?
Bounded rationality	Limited time, memory, and calculation	Where should I spend attention?

A crucial distinction

Chess is a perfect-information game because the board and legal history are observable. Human chess is still a decision problem under uncertainty because calculation, memory, intentions, and future errors are not observable. Do not confuse hidden minds with hidden pieces.

Contents

Preface: The move is not the point	ii
A map from game theory to chess	iv
I Model the Decision	1
1 What kind of game is chess?	2
1.1 The formal answer	2
1.2 The human answer	3
1.3 The five-step loop	3
1.4 A first equilibrium intuition	4
2 Rules are the game frame	6
2.1 Skeleton before personality	6
2.2 Why tiny rules change large strategies	7
2.3 Rules refresher as strategic language	8
2.4 The state test	8
3 States, actions, outcomes, and strategies	10
3.1 Four objects, four different jobs	10
3.2 The tree, not the line	10
3.3 The candidate menu	11
3.4 Information sets and human blindness	12
3.5 Cutting the tree honestly	12
4 Preferences and practical utility	14
4.1 Utility ranks outcomes	14
4.2 Four layers of utility	14
4.3 Risk attitude changes with context	15

4.4	Rational does not mean timid or selfish	15
4.5	A practical utility card	16
5	Evaluation without a magic number	18
5.1	Evaluation is compression	18
5.2	Seven questions	18
5.3	Nominal value and functional value	19
5.4	From feature to plan	19
5.5	Option value and irreversibility	20
II	Reason Through the Tree	22
6	Best responses and candidate moves	23
6.1	A move is good relative to a reply	23
6.2	Generate from two viewpoints	23
6.3	The reply-first filter	24
6.4	Counterfactual candidate generation	24
6.5	Best response does not mean engine move	25
7	Dominance and disciplined elimination	26
7.1	Dominance in plain language	26
7.2	Why chess resists easy dominance	26
7.3	Four safe elimination tests	27
7.4	Iterated elimination as calculation	27
7.5	A dominance ledger	28
8	Minimax and the equilibrium compass	29
8.1	The adversarial value	29
8.2	Security level	29
8.3	Equilibrium as mutual best response	30
8.4	When minimax is the right lens	30
8.5	Robust moves	31
9	Backward induction and tactical proof	32

9.1	Start at the end	32
9.2	Threat, tactic, and force	33
9.3	The forcing ladder	33
9.4	Induction over a line	34
9.5	Known end states as libraries	34
10	Dynamic value: tempo, initiative, and zugzwang	36
10.1	A move spends a turn	36
10.2	Development, initiative, and force	36
10.3	The initiative ledger	37
10.4	Move-order economics	37
10.5	Zugzwang: when action has negative value	38
10.6	Convert or renew	38
III	Choose Under Risk and Limits	40
11	Search under bounded rationality	41
11.1	Rationality with a budget	41
11.2	Search where value can flip	41
11.3	Selective deepening	42
11.4	Pruning without lying	42
11.5	Horizon and quiescence	43
11.6	Chunking and pattern libraries	43
12	Expected utility and practical chances	45
12.1	From worst case to weighted branches	45
12.2	Probability is conditional	45
12.3	Objective floor, practical ceiling	46
12.4	Error asymmetry	46
12.5	A simple decision table	47
12.6	Risk dominance in practical chess	47
13	Mixed strategies and the value of unpredictability	49

13.1 Pure and mixed strategies	49
13.2 Unpredictability is not randomness for its own sake	49
13.3 The exploitability test	50
13.4 Mixing at different levels	50
13.5 When not to mix	50
14 Time controls and clock utility	53
14.1 The clock changes the game	53
14.2 Think in decisions, not moves	53
14.3 The time budget equation	54
14.4 Stop rules	54
14.5 Low-time protocol	55
14.6 Clock as signal and weapon	55
IV Think About Minds	57
15 Beliefs and Bayesian updating	58
15.1 The board state and the player type	58
15.2 Strong and weak evidence	58
15.3 Priors should be broad, not prejudiced	59
15.4 Bayesian candidate choice	59
15.5 Learning from time usage	60
15.6 The belief ledger	60
16 Signals, threats, and credible commitment	62
16.1 Moves do two jobs	62
16.2 Cheap talk and costly signals	62
16.3 Threats as incentive design	63
16.4 Credibility test	63
16.5 Commitment devices	64
16.6 Inducing choices	64
17 Prophylaxis and common knowledge	66

17.1	Thinking one mind deeper	66
17.2	Common knowledge	66
17.3	Four responses to an opponent's plan	67
17.4	The free-move experiment	67
17.5	Useful restriction	68
17.6	Mutual plans as a matrix	68
18	Biases, emotions, and distorted utility	70
18.1	Bounded rationality is systematic	70
18.2	Eight recurring biases	70
18.3	Emotion as data, not command	71
18.4	Tilt as preference corruption	71
18.5	The narrative audit	72
19	Repeated encounters, reputation, and adaptation	73
19.1	One game changes the next	73
19.2	Reputation as a commitment asset	73
19.3	Exploit, explore, or conceal	74
19.4	Adaptation loops	74
19.5	Do not overfit one game	74
19.6	Tournament ecology	75
V	Apply the Framework	76
20	The opening as a prior, not a script	77
20.1	A prior narrows search	77
20.2	Opening goals as option creation	77
20.3	Move orders as information games	78
20.4	Preparation as a contingent strategy	78
20.5	Leaving book well	79
20.6	Repertoire economics	79
21	The middlegame as a portfolio of imbalances	81

21.1	An imbalance is a persistent asymmetry	81
21.2	Plan portfolios	82
21.3	Role assignment	82
21.4	Create a second weakness	83
21.5	Pawn breaks as state transitions	83
21.6	Plan failure and switching	84
22	Sacrifice as intertemporal exchange	85
22.1	Material now for resources later	85
22.2	Compensation categories	85
22.3	Three kinds of sacrifice	86
22.4	The three-column proof	86
22.5	Time decay of compensation	87
22.6	Return sacrifice	87
23	Endgames as compressed game trees	89
23.1	Fewer pieces, sharper state values	89
23.2	The endgame evaluation order	89
23.3	The king as a production asset	90
23.4	Passed pawns and option value	90
23.5	Exchange decisions point to terminal states	90
23.6	Known positions as value functions	91
23.7	Practical endgame utility	91
24	Match, tournament, and draw strategy	93
24.1	The event changes preferences	93
24.2	Risk profiles by need	94
24.3	Draw offers as sequential choices	94
24.4	Color and asymmetric utility	94
24.5	Match games and dynamic strategy	95
24.6	Ethics and rule compliance	95

VI Practice Until It Is Automatic	97
25 The post-game learning loop	98
25.1 A result is a noisy label	98
25.2 The five-pass review	98
25.3 Error taxonomy	99
25.4 Counterfactual discipline	99
25.5 Using engines without outsourcing thought	100
25.6 Pattern cards and spaced retrieval	100
25.7 Process metrics	101
26 A twelve-week course in thinking	102
26.1 The training principle	102
26.2 Weekly plan	102
26.3 A 45-minute study session	103
26.4 Skill ladders	103
26.5 The integrated move protocol	104
26.6 Assessment positions	104
26.7 Retention schedule	105
26.8 Graduation standard	105
VII Reference and Solutions	107
A Notation and rules reference	108
A.1 Board coordinates	108
A.2 Piece symbols and names	108
A.3 Algebraic notation	109
A.4 Essential rule reminders	109
A.5 Nominal material values	110
A.6 A compact analysis notation	110
B Decision sheets and printable templates	111
B.1 Pre-game utility card	111

B.2	BOARD position sheet	111
B.3	Calculation tree sheet	113
B.4	Post-game five-pass sheet	113
B.5	Pattern card	114
C	Solutions and answer guidance	115
D	Glossary of chess decision terms	123
	Conclusion: See the reply	129
	One-page field guide	130
	Method and source note	131

List of Figures

2.1	The initial board is only a frame. Preferences, clocks, and context turn it into a practical decision problem.	7
3.1	Calculation must branch at the opponent's decision.	11
8.1	Minimax compares A's worst leaf -5 with B's worst leaf 0 , so it chooses B.	30
9.1	White to move can use the terminal target of back-rank mate to reason backward.	33
23.1	In simplified endings, the same placement can change value when the side to move changes. Treat the move as part of the state.	91
A.1	Coordinates and the initial position, viewed from White's side.	108

List of Tables

Part I

Model the Decision

What kind of game is chess?

You will be able to describe chess as a sequential, deterministic, two-player, largely zero-sum game of perfect information; separate the formal game from the human problem; and use the MODEL–REPLY–VALUE–DECIDE–UPDATE loop.

The question underneath the chapter

Why can a game with no hidden pieces still feel uncertain?

1.1 The formal answer

A chess position is a decision state. One player chooses a legal move, the board changes, and the other player chooses. The process continues until checkmate, resignation, or a draw rule ends the game. Nothing random moves a piece. Both players see the same board. If we assign utilities $+1$ to a win, 0 to a draw, and -1 to a loss, one player's gain is the other's loss.

That description gives chess four formal properties:

Sequential.

Players act one after another, so order matters.

Deterministic.

A move has a fixed legal consequence; there are no dice.

Perfect information.

The relevant board state and prior moves are observable.

Constant-sum at the result level.

A win for one side is a loss for the other, while a draw gives neither side the win.

Formal lens

Let s denote a position and $A(s)$ its legal moves. A move $a \in A(s)$ produces a new state $T(s, a)$. A complete strategy is not one move; it is a rule that specifies an action at every state the player might face.
For a terminal state z , a simple result utility for White is

$$u_W(z) \in \{-1, 0, +1\}, \quad u_B(z) = -u_W(z).$$

1.2 The human answer

If chess were only the formal game, the entire problem would be “play perfectly.” Human beings cannot inspect the full tree. We have clocks, incomplete memories, uneven calculation, emotions, opening knowledge, and beliefs about one another. The board is public; the computations are private.

This creates two layers:

	Formal chess	Practical chess
Uncertainty	None about the board’s transition	Much about future choices and errors
Goal	Maximize game-theoretic result	Maximize result under time and cognitive limits
Opponent	An optimal responder	A particular person with tendencies
Move quality	Value under best play	Value under best play plus usability

A move can therefore be objectively best yet practically poor for you. Imagine two continuations. Move A preserves a tiny advantage but requires ten precise only-moves. Move B gives an equal position with a clear plan and little danger. In an analysis room, A may rank first. With thirty seconds on the clock, B may maximize your actual chance of scoring.

Intuition

The board tells you what is true. The clock and the players tell you what is usable.

1.3 The five-step loop

Every chapter elaborates one stage of the same loop.

MODEL. State the facts before inventing a move. Count material. Locate loose pieces. Compare king safety, activity, pawn structure, space, and time. Name whose move it is and what changed.

REPLY. Treat the opponent as an active decision maker. For each serious candidate, ask for the most annoying legal response. This single habit cures much “hope chess.”

VALUE. Compare the resulting branches, not the beauty of the first move. A sacrifice is not valuable because it looks brave; it is valuable if its continuation yields sufficient utility.

DECIDE. Choose using the right standard. In a forcing tactical line, protect against the strongest reply. In a quiet practical choice, include complexity, time, and opponent tendencies.

UPDATE. After every move, rebuild the model. A move is evidence. It changes the board and may reveal preparation, fear, misunderstanding, or a plan.

Worked example: the attractive move

You see a queen move that attacks h7. **MODEL:** your back rank is weak and your queen currently defends a rook. **REPLY:** after the queen leaves, the opponent has a forcing rook invasion. **VALUE:** your attack is a possibility; the rook loss is immediate. **DECIDE:** reject the queen move. **UPDATE:** search for an attacking move that preserves the defensive function.

The lesson is not “never attack h7.” It is “evaluate a move as a bundle of changed relationships.”

1.4 A first equilibrium intuition

An equilibrium is a pair of strategies from which neither player benefits by changing alone. In chess, this idea is less a position you can point to than a standard of mutual resistance: your move must make sense after the opponent replies well, not merely after the reply you hope to see.

When a puzzle says “White to move and win,” it asserts that White has a strategy that forces a winning terminal outcome against every defense. The solution is not the first move alone. It is the complete contingent plan.

Memory hook

Move quality lives one reply later. Before approving a candidate, say aloud: “And then what is their best response?”

Exercise 1.1: Classify the game

Explain why chess can be perfect-information and still require beliefs about an opponent. Give one board fact and one hidden human fact.

Exercise 1.2: Use the loop

Choose any recent blunder. Rewrite the decision using MODEL, REPLY, VALUE, DECIDE, UPDATE. At which stage did the process fail?

The chapter in one minute

Formal chess has no hidden board state and no chance moves. Practical chess is uncertain because people cannot search the full tree. Your universal loop is MODEL, REPLY, VALUE, DECIDE, UPDATE.

Rules are the game frame

You will distinguish a game frame from preferences; recognize which details belong in the state; and understand why legal moves, clocks, draw rules, and side to move are strategic facts rather than bookkeeping.

The question underneath the chapter

What must be specified before “the best move” is even a meaningful phrase?

2.1 Skeleton before personality

A **game frame** tells us who acts, what actions are available, how actions change the state, and what terminal outcomes can occur. It does not yet tell us how a player ranks those outcomes. In chess, the frame includes more than the placement of pieces.

A complete practical frame records:

- piece locations and side to move;
- castling and en-passant rights;
- repetition and move-count history when relevant;
- legal end conditions and scoring rules;
- clock times, increment or delay, and time-control stage;
- match context if it changes the value of a draw or loss.

A diagram without side to move is not a complete decision problem. The same arrangement can be winning for one side to move and losing for the other because a forcing action or zugzwang is available now.

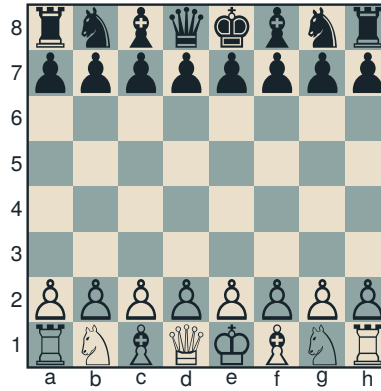


Figure 2.1: The initial board is only a frame. Preferences, clocks, and context turn it into a practical decision problem.

2.2 Why tiny rules change large strategies

Rules determine incentives. Consider three settings:

- a) a casual game where a draw is acceptable;
- b) the last round of a tournament where only a win earns a prize;
- c) a match where the player with White advances with a draw.

The legal board can be identical while the preferred continuation changes. In the second setting, a safe forced draw may have almost no practical utility. In the third, taking unnecessary risk may be irrational even if an engine reports a small advantage.

Similarly, a five-second increment rewards moves that preserve clock solvency. A sudden-death finish makes a technically winning but mechanically difficult ending less attractive. The rules do not merely surround the game; they shape the utility of branches.

Formal lens

A frame can be written as

$$\mathcal{F} = (N, S, A, T, Z),$$

where N is the set of players, S the states, $A(s)$ the legal actions at state s , T the transition rule, and Z the terminal states. Adding each player's preference ordering over Z turns the frame into a game.

2.3 Rules refresher as strategic language

This is not a full beginner’s manual, but four rules deserve strategic emphasis.

Check restricts choice. A checked player may choose only moves that remove check. This shrinks the action set and explains why checks are forcing.

Pins restrict effective choice. A pinned piece may be legally or practically unable to move. A legal action that loses the king or queen has low utility, so the effective action set can be much smaller than the legal set.

Promotion changes future utility. A pawn near promotion is not “worth one” in the ordinary sense. Its option to transform makes its state-dependent value much larger.

Draw rules create terminal exits. Repetition, stalemate, insufficient mating material, and move-count rules can convert a lost material position into a drawable game. They are resources, not technical footnotes.

Worked example: side to move is part of the state

Suppose each side has a king and White has one pawn. The piece placement alone does not determine the result. If the side to move must yield opposition, the value can flip. Therefore a position is not a photograph; it is a photograph plus the right to act.

2.4 The state test

Before analysis, ask whether a missing fact could change the legal moves or your ranking of outcomes. If yes, it belongs in the model.

Missing fact	Possible consequence
Side to move	Tactic or zugzwang reverses
Castling rights	King-safety plan changes
Last pawn move	En-passant may exist
Repetition count	A draw claim may be available
Clock time	Complexity changes practical value
Tournament need	Draw utility changes

Common error

Do not analyze an under-specified diagram as though it were a full game. Ask: whose move, what rights, what clock, what score?

Exercise 2.1: Frame or preference?

Classify each item as part of the frame or a preference: White has 42 seconds; White dislikes queenless endings; castling rights are lost; a draw wins the match; Black enjoys complications.

Exercise 2.2: Missing state

Find a chess diagram online or in a book. List four facts you would need before recommending a practical move in a tournament.

The chapter in one minute

The board is not the whole game. The frame includes legal rights, history, clocks, rules, and context. The game begins when preferences rank the possible outcomes.

States, actions, outcomes, and strategies

You will separate a move from a strategy, read chess as an extensive-form tree, and build candidate sets small enough for a human to analyze.

The question underneath the chapter

Why is “I plan to attack” not yet a strategy?

3.1 Four objects, four different jobs

Chess language often mixes terms that game theory keeps separate.

State.

Everything currently relevant: board, rights, clocks, history, context.

Action.

One legal move chosen now.

Outcome.

A terminal result, or a future position used as a temporary evaluation point.

Strategy.

A contingent rule for what to do at every state you might reach.

“Play on the kingside” is an intention. “After . . . h6, retreat the bishop to h4; after . . . g6, prepare h5; after central counterplay, close the center only if it does not lose d5” begins to resemble a strategy because it specifies responses to alternatives.

3.2 The tree, not the line

A common calculation error is to imagine one attractive continuation. A game tree branches because the opponent chooses too.

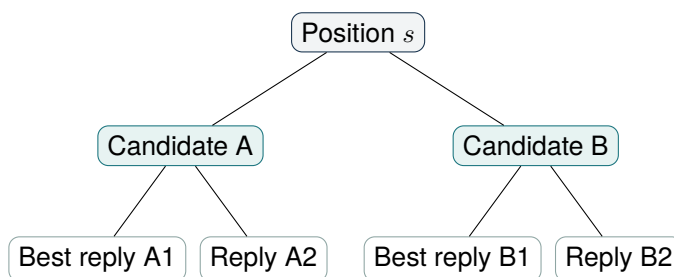


Figure 3.1: Calculation must branch at the opponent's decision.

Humans cannot expand every legal move. The first skill is not calculation depth; it is **candidate discipline**. Build a compact menu that covers the important strategic functions.

3.3 The candidate menu

Use CCT+I:

- C** Checks: forcing because the reply set is restricted.
- C** Captures: alter material and lines immediately.
- T** Threats: create a future loss the opponent must address.
- I** Improvements: develop the worst piece, prevent a break, improve structure, or create an option.

CCT is a search order, not a command to play violently. Most positions are decided by quiet improvements. The purpose is to avoid missing forcing resources before spending time on plans.

A robust candidate set usually contains three to five moves:

1. the most forcing move;
2. the most natural positional move;
3. the opponent's feared move played by you, if legal;
4. one move that changes the pawn structure;
5. one defensive or simplifying move when under pressure.

3.4 Information sets and human blindness

In formal chess, each player knows the exact state, so every decision node is its own information set. In human thought, attention creates a shadow version of incomplete information: you may fail to notice that a square is defended or that a tactic exists. The fact is visible but not represented in your mental model.

That distinction matters. A surprise tactic is not random. It was a branch you failed to include. The cure is not “be luckier” but “improve state reconstruction and candidate generation.”

Worked example: plan versus strategy

Plan: “Trade queens because I am ahead.”

Contingent strategy: “Offer the queen trade if it preserves the extra pawn and prevents counterplay. If the opponent avoids the trade by moving the queen to an active square, use the gained tempo to double rooks. If the trade enters a pawn ending where the king is cut off, keep queens instead.”

The strategy survives different replies; the slogan does not.

3.5 Cutting the tree honestly

Every analysis stops before checkmate. A **horizon** is the leaf at which you stop and evaluate. Good calculation stops at a quiet, understandable position: the forcing sequence is over, loose pieces are accounted for, and the next plan can be described. Stopping immediately after winning a pawn is dangerous if the opponent then traps your queen.

Memory hook

A line is what you hope happens. A tree is what the opponent is allowed to choose.

Exercise 3.1: Action or strategy?

Rewrite “I will attack the isolated pawn” as a contingent strategy containing at least three possible opponent responses.

Exercise 3.2: Build candidates

From any middlegame position, list one check, one capture, one threat, and two improving moves. Eliminate illegal or obviously losing moves only after listing them.

The chapter in one minute

A move is one action; a strategy is a complete contingent plan. Search a tree, not a fantasy line. Generate candidates with CCT+I and stop calculation only at a stable horizon.

Preferences and practical utility

You will understand utility as a ranking rather than money; distinguish objective result utility from practical utility; and choose risk according to score, clock, and personal error profile.

The question underneath the chapter

When can a draw be more valuable than a small objective advantage?

4.1 Utility ranks outcomes

Utility numbers are labels that preserve preferences. If you assign 10 to a win, 6 to a draw, and 0 to a loss in a match situation, the numbers say a draw is preferred to a loss and a win to a draw. They do not prove that a win creates exactly ten units of happiness.

For pure chess result, the standard ranking is

$$\text{win} \succ \text{draw} \succ \text{loss}.$$

But practical decisions often include more detail. A player may prefer an easy draw to a technically equal but dangerous ending, or a complex equal position to a sterile one when only a win matters.

Intuition

Utility is a scoreboard for choices, not a thermometer for feelings.

4.2 Four layers of utility

Use four nested layers rather than one magic evaluation.

1. **Result utility:** win, draw, or loss.

2. **Event utility:** match advancement, tournament points, tie-breaks, prize goals.
3. **Practical utility:** ease, clock demand, error asymmetry, familiarity.
4. **Learning utility:** whether a choice helps develop skill when result is not the only goal.

These layers can conflict. In training, entering a difficult rook ending may have high learning utility. In a must-draw final round, the same choice may have poor event utility.

4.3 Risk attitude changes with context

Suppose two moves have these rough practical distributions:

	Win	Draw	Loss
Move A: sharp	45%	20%	35%
Move B: stable	25%	65%	10%

If standard scoring gives 1 for a win, $\frac{1}{2}$ for a draw, and 0 for a loss, the rough expected scores are

$$EU(A) = .45 + .5(.20) = .55, \quad EU(B) = .25 + .5(.65) = .575.$$

B is slightly better by expected score. But if a draw eliminates you and only a win matters, the win probability may dominate; A becomes attractive. If a draw guarantees first place, B is strongly preferred.

The percentages are estimates, not objective facts. Their value lies in forcing you to make the tradeoff explicit.

4.4 Rational does not mean timid or selfish

A sacrifice can be rational. A draw offer can be rational. Playing a quiet move against an attacking player can be rational. Rationality means choosing consistently with your preferences and beliefs, not following a universal personality.

Problems arise when momentary emotion silently rewrites utility. After missing a win, players often reject a draw because they feel “entitled” to the earlier advantage. The past position is gone. Utility should be assigned to current future outcomes, not to recovering a vanished story.

Sunk-position fallacy

Do not take extra risk merely to get back the advantage you had ten moves ago. The current state is the only launch point for future utility.

4.5 A practical utility card

Before a critical game, write one sentence for each:

- **Result need:** “A draw is enough” or “I need winning chances.”
- **Clock preference:** “Avoid a line requiring repeated only-moves.”
- **Position preference:** “I understand closed structures better than open king hunts.”
- **Emotional guardrail:** “After a mistake, I will rebuild the model before changing risk.”

This is not an excuse to avoid objective chess. It is a way to translate objective options into the utility relevant to this game.

Worked example: equal is not identical

Two continuations are evaluated as equal. One leads to opposite-side castling with attacks against both kings. The other trades queens into a symmetrical minor-piece ending. The formal result value may be equal under perfect play. Their practical utilities differ by clock, skill, event need, and opponent type. “Equal” does not mean interchangeable.

Exercise 4.1: Rank outcomes

Create utility rankings for three contexts: a casual training game, a must-draw match game, and a must-win final round. Include at least four outcomes, such as easy draw, difficult draw, sharp winning chance, and low-risk edge.

Exercise 4.2: Detect emotional utility

Describe a time anger, fear, or pride changed the value you assigned to an option. What would a pre-game guardrail have said?

The chapter in one minute

Utility ranks what matters now. Separate result, event, practical, and learning utility. Context changes rational risk; emotion should not secretly change it

after the fact.

Evaluation without a magic number

You will evaluate a position as a vector of resources, understand context-dependent piece value, and convert static features into plans and candidate moves.

The question underneath the chapter

What does an engine's "+0.8" hide?

5.1 Evaluation is compression

The full future tree is too large, so players compress a position into features. An engine number is one summary of the estimated future result under strong play. Human evaluation should begin with a vector:

$$E(s) = (M, K, A, P, S, T, O),$$

where M is material, K king safety, A activity, P pawn structure, S space, T time or initiative, and O option value.

The components interact. An extra pawn matters less if your king is being mated. A bishop matters more when the center opens. Space matters when you can use it without creating targets. Option value matters because a flexible move preserves several future actions.

5.2 Seven questions

1. **Material:** What is present, attacked, loose, overloaded, or trapped?
2. **King safety:** Which checks, open lines, weak colors, and defenders matter?
3. **Activity:** Which piece has the fewest useful squares? Which piece invades?

- 4. **Pawn structure:** What fixed weaknesses, passed pawns, breaks, and chains exist?
- 5. **Space:** Who controls useful territory, and can the cramped side break free?
- 6. **Time:** Who is making threats? Who must respond?
- 7. **Options:** Which irreversible choices can be delayed? Which moves keep plans alive?

Answering all seven prevents one-dimensional thinking. “I am up a pawn” is not an evaluation. It is one coordinate.

5.3 Nominal value and functional value

The familiar scale—pawn 1, knight 3, bishop 3, rook 5, queen 9—is a starting prior. Functional value depends on the state.

Piece	Value rises when	Value falls when
Knight	stable outpost, closed center, local targets	board is open, no central squares
Bishop	open diagonals, targets on both wings	own pawns block it, wrong-color duties
Rook	open file, seventh rank, active king cut-off	no entry squares, defensive passivity
Queen	exposed kings, many coordinated threats	vulnerable to tempo, simplification favored
Pawn	passed, protected, near promotion, space-gaining	fixed, isolated, blockaded, overextended

A trade should therefore compare future functions, not labels. Exchanging a “three-point bishop” for a “three-point knight” can be excellent or disastrous depending on the remaining pawn structure and squares.

5.4 From feature to plan

An evaluation is useful only if it generates actions. Translate nouns into verbs:

Feature	Planning verbs
Unsafe enemy king	open, remove defender, bring attacker, restrict flight
Weak pawn	fix, attack, overload defender, create second weakness
Bad piece	reroute, exchange, free with pawn break
Space advantage	restrict, improve, switch wings, avoid premature release
Passed pawn	support, advance, distract, clear promotion square
Opponent initiative	trade attackers, return material, close lines, create counter-threat

Worked example: a pawn is not a plan

You have an extra queenside pawn but your rook is passive and the opponent's pieces surround your king. Static phrase: "up a pawn." Vector evaluation: $M+$, $K--$, $A-$, $P+$, $T-$, O uncertain. Plan: neutralize checks, offer a queen trade, activate the rook even at the cost of returning the pawn. The material edge is a resource that can be spent to improve the more urgent coordinates.

5.5 Option value and irreversibility

Pawn moves, exchanges, and king commitments often remove future choices. A move with slightly less immediate gain may be superior because it preserves multiple plans. Call this **option value**.

A useful test is:

If my opponent waits, what plans remain after this move? What plans disappear forever?

Flexible does not mean passive. A forcing move may have enormous value. The point is to charge irreversible actions for the options they destroy.

Formal lens

A practical evaluation can be viewed as a weighted function

$$U(s) = w_M M + w_K K + w_A A + w_P P + w_S S + w_T T + w_O O,$$

but the weights w_i depend on the position. In a mating attack, w_K becomes

dominant. In a quiet pawn ending, pawn structure and king activity dominate. The formula is a reminder of dependence, not a calculator.

Exercise 5.1: Vector evaluation

Choose a current position from one of your games. Rate each component *M, K, A, P, S, T, O* as favorable, equal, or unfavorable. Name one candidate produced by each unfavorable component.

Exercise 5.2: Functional trade

Find an exchange from a recent game. Compare the pieces before and after using mobility, targets, defensive duties, and future pawn structure rather than nominal values alone.

The chapter in one minute

Evaluation is a compressed vector, not a single adjective. Inspect material, king safety, activity, structure, space, time, and options. Then turn features into verbs and candidates.

Part II

Reason Through the Tree

Best responses and candidate moves

You will define a best response, generate candidates from both sides of the board, and use a reply-first test to prevent one-move blunders.

The question underneath the chapter

What move would you choose if your opponent were allowed to answer well?

6.1 A move is good relative to a reply

In game theory, a **best response** is an action that gives a player the highest utility against a specified action or strategy of the other player. In chess, you usually do not know the opponent's complete strategy, but you can ask a local version:

After my candidate, which legal reply gives the opponent the best resulting position?

The answer is often not the move you expect psychologically. It may be a forcing check, an in-between move, a quiet defense, a counterattack, or a simplifying exchange.

6.2 Generate from two viewpoints

Most players generate candidates only for themselves. Strong practical reasoning alternates perspectives.

Pass 1: My forcing resources. Checks, captures, direct threats.

Pass 2: Their forcing resources. Checks, captures, direct threats after each candidate.

Pass 3: My structural improvements. Worst piece, break, restriction, simplification.

Pass 4: Their strategic goal. What would they play if granted a free move?

The fourth pass is especially powerful. If the opponent wants . . . d5, candidates that prevent or punish . . . d5 gain value. If they want to trade queens, you can decide whether that preference aligns with yours.

Intuition

Do not ask only “What can I do?” Ask “What are they trying to make true?”

6.3 The reply-first filter

Before calculating deeply, run every serious candidate through **SAFE**:

- S** **Scan checks** against your king.
- A** **Assess captures** of your loose or newly undefended pieces.
- F** **Find threats** the move permits or creates for the opponent.
- E** **Evaluate the end** of the forcing sequence, not its first pleasant moment.

SAFE is a tactical hygiene routine. It should take seconds in quiet positions and more time when kings or queens are exposed.

6.4 Counterfactual candidate generation

A useful candidate is sometimes discovered by reversing roles:

If I could move for my opponent now, what would I play?

That move identifies the opponent’s best response to your pass. You can then search for candidates that make it impossible, harmless, or costly. This is the heart of prophylaxis, developed later.

Worked example: the natural recapture

Your opponent captures a pawn. The automatic response is to recapture. Before doing so, generate from both viewpoints. Your forcing resource is the recapture. Their forcing resource after it is a queen check that wins your rook. Your alternative is an in-between check that forces the king away, then recaptures safely. The best response test turns “obvious” into “verified.”

6.5 Best response does not mean engine move

At the board, you estimate the best response with limited time. The goal is not certainty; it is disciplined adversarial imagination. Use three levels:

1. **Forced:** the reply is legally required or nearly so.
2. **Natural:** it follows a visible principle and is likely to be found.
3. **Difficult:** it is strong but hidden, counterintuitive, or precise.

For objective verification, test all three. For practical choice, their findability matters too. Never label a defense “unlikely” merely because it refutes your idea; first decide whether it is actually difficult.

Memory hook

A candidate earns calculation only after it survives one hostile glance.

Exercise 6.1: Opponent’s free move

Take a position from your latest game. Give the opponent a hypothetical free move. What would they choose? Generate two candidates that address that goal in different ways.

Exercise 6.2: SAFE audit

For one move you considered but rejected, run SAFE explicitly. Which category supplied the refutation or confirmation?

The chapter in one minute

A best response maximizes the opponent’s utility against your candidate. Generate from both viewpoints, use the opponent’s free move to reveal goals, and run SAFE before deep calculation.

Dominance and disciplined elimination

You will distinguish strict and weak dominance, understand why dominance is rare in chess, and eliminate candidates without discarding hidden resources.

The question underneath the chapter

When is it safe to stop analyzing a move?

7.1 Dominance in plain language

Move A **strictly dominates** Move B over a set of relevant opponent replies if A produces a better result than B against every reply in that set. A **weakly dominates** B if A is never worse and is better against at least one reply.

In a small payoff table:

	Opponent defends	Opponent counterattacks
Move A	+	=
Move B	=	–

A is at least as good in both columns and better in each comparison here, so B can be removed from this simplified decision.

7.2 Why chess resists easy dominance

Chess continuations are vast. A move that looks inferior against two obvious replies may contain a resource against a third. Positional value is also multidimensional: one move gains material, another gains safety, and their ranking can reverse later.

Therefore use dominance as a disciplined pruning rule, not a slogan. State the comparison domain:

“Against the three forcing replies I can identify, A reaches the same favorable structure as B while keeping an extra tempo.”

That is stronger than “A looks more active.”

7.3 Four safe elimination tests

Legality elimination. Remove moves that leave your king in check or violate the rules. Obvious, but under time pressure illegal mental branches consume attention.

Forcing refutation. Remove a candidate if one clear opponent reply forces unacceptable utility and you have no counter-resource.

Transposition dominance. If A and B reach the same position but A gives the opponent an extra useful tempo, B may dominate A.

Function coverage. If two moves perform the same job, prefer the one that also preserves another function: defense, flexibility, development, or clock ease.

Worked example: same square, different order

You can prepare a rook lift immediately or first play a useful king-safety move. Both routes reach the rook-lift position, but the immediate route allows a forcing countercheck. The improved move order covers the same attacking function while removing the countercheck. Over the relevant replies, the safer order dominates.

7.4 Iterated elimination as calculation

Game theory sometimes eliminates dominated strategies repeatedly. Chess calculation often does the same:

1. Generate four candidates.
2. Remove one that loses to a forcing check.
3. After inspecting replies, remove another that transposes a tempo down.
4. Compare the remaining two deeply.

This is not laziness. It is rational allocation of a limited search budget. The danger is premature elimination based on aesthetics: “knights belong in the center,” “never

move the same piece twice,” or “queen trades favor the defender.” Principles suggest candidates; concrete replies decide.

The invisible-column error

Dominance claims are only as good as the opponent replies you included. Before deleting a candidate, ask whether a forcing or quiet resource creates a new column in the comparison table.

7.5 A dominance ledger

When the choice is difficult, write or think in three rows:

Candidate A		Candidate B
Shared function	What both accomplish	What both accomplish
Extra benefit	Preserved option or gain	Preserved option or gain
Unique risk	Strongest refutation	Strongest refutation

If one move matches the shared function, has an extra benefit, and no larger unique risk, you have a practical dominance argument.

Exercise 7.1: Dominance statement

Compare two plausible moves from a game using at least three opponent reply classes. State whether one dominates the other or whether the ranking depends on the reply.

Exercise 7.2: Recover a deleted move

Find a puzzle or game where an odd-looking move is best. Which principle might have caused premature elimination, and what hidden reply makes the move work?

The chapter in one minute

Dominance lets you prune, but only over replies you have honestly considered. Eliminate by legality, forcing refutation, inferior transposition, or function coverage—not by taste.

Minimax and the equilibrium compass

You will use minimax as a standard of resistance, understand its relation to equilibrium, and know when worst-case reasoning should override hopeful expected-value reasoning.

The question underneath the chapter

What remains of your idea after the opponent chooses the branch you like least?

8.1 The adversarial value

Minimax says: choose the action whose worst credible continuation is best. If White is maximizing and Black minimizing White's utility, then schematically

$$V(s) = \max_{a \in A_W(s)} \min_{b \in A_B(T(s,a))} V(T(T(s,a), b)).$$

The formula is not asking you to calculate the whole game. It disciplines the comparison: each candidate is judged by the opponent's strongest reply, not the average reply or the reply that makes your combination beautiful.

A convenient engine-style form is **negamax**:

$$V(s) = \max_{a \in A(s)} [-V(T(s, a))],$$

where value is always measured from the player-to-move's perspective.

8.2 Security level

The worst outcome your strategy can guarantee is its **security level**. In a tactical position, this is essential. If Move A wins a queen after a natural defense but loses to one forced resource, its security level is a loss. Move B that forces a draw has a

higher security level when a draw is acceptable.

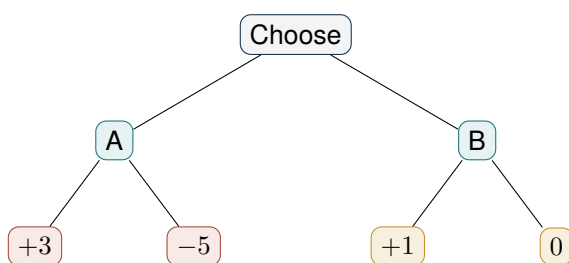


Figure 8.1: Minimax compares A's worst leaf -5 with B's worst leaf 0 , so it chooses B.

8.3 Equilibrium as mutual best response

A Nash equilibrium is a pair of strategies in which neither player can improve by changing strategy alone. In finite deterministic perfect-information zero-sum games, backward reasoning assigns the game a value and optimal pure strategies exist. For chess, the exact value of the initial position is not known to us, but the concept still supplies a compass: a sound plan must survive optimal resistance.

Do not confuse equilibrium with a drawish position. An equilibrium strategy may force a win if the game-theoretic value is a win. Equilibrium means mutual optimality, not equality on the board.

8.4 When minimax is the right lens

Use worst-case reasoning heavily when:

- the line is forcing and replies are few;
- one opponent reply changes the result category;
- you are defending and cannot permit a tactical failure;
- a draw or win can be forced;
- the opponent has enough time and skill to find the resource.

Use it less rigidly when comparing several objectively equal, quiet positions against a fallible human. There, practical expected utility and complexity matter. Chapter 12 combines the lenses.

8.5 Robust moves

A **robust** move performs reasonably well across many opponent responses and across small errors in your own evaluation. Robustness has three forms:

Reply robustness.

No single reply collapses the position.

Evaluation robustness.

The move remains sensible if one feature was misjudged.

Execution robustness.

The continuation contains several acceptable moves, not repeated only-moves.

Robust does not mean passive. A forcing tactical win is maximally reply-robust if every defense fails.

Worked example: the sacrifice test

You consider a rook sacrifice. Against the capture you calculated mate. Minimax asks the defender to choose the reply that minimizes your utility. Can the opponent decline the rook? Interpose a piece? Give a countercheck? Trade queens? If one branch leaves you simply down a rook, the sacrifice is not forced. The best-response search is the proof obligation.

Memory hook

For forcing decisions, do not average the branches until you know none of them contains a cliff.

Exercise 8.1: Security levels

Invent or find a decision with two candidates. Assign rough values to each candidate's two strongest replies. Which move has the better security level? Does that match your intuition?

Exercise 8.2: Robustness audit

Evaluate a chosen move for reply, evaluation, and execution robustness. Which form was weakest?

The chapter in one minute

Minimax chooses the best worst case. Equilibrium is mutual best response, not necessarily a draw. Use minimax for forcing branches and result-changing risks; favor robust moves when uncertainty is high.

Backward induction and tactical proof

You will solve forcing lines from terminal or stable leaves backward, distinguish threats from forced outcomes, and use proof standards for combinations.

The question underneath the chapter

Why is it easier to calculate when you know what the final position must look like?

9.1 Start at the end

Backward induction solves a sequential game by evaluating the last decisions first. In chess, tactics become manageable when you identify terminal targets:

- checkmate;
- a forced draw;
- a won queen or decisive material;
- promotion;
- a stable ending with a known result;
- a quiet leaf whose evaluation is clear.

Instead of asking “What sequence looks strong?” ask “What final state would prove the claim, and which forcing moves make it unavoidable?”

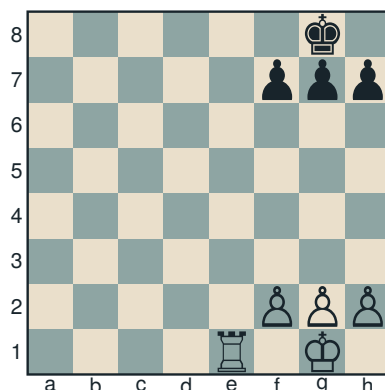


Figure 9.1: White to move can use the terminal target of back-rank mate to reason backward.

In the diagram, **Re8#** is not persuasive because the rook looks active. It is persuasive because the leaf is checkmate. Work backward: the rook on e8 checks along the eighth rank; the king has no legal flight square; no piece can capture or block. The terminal proof validates the first move.

9.2 Threat, tactic, and force

A **threat** says, “If you do nothing, I will obtain a favorable outcome.” A **forced line** says, “Every legal or relevant response still leads to the outcome.” Confusing them produces false combinations.

Use three proof questions:

1. What exactly is the terminal claim?
2. What are all defensive categories: move king, capture attacker, block, countercheck, counter-threat, decline sacrifice?
3. At the leaf, is the position quiet enough to evaluate?

9.3 The forcing ladder

Calculation is easiest when actions shrink the opponent’s reply set. A useful order is:

mate threats → checks → forcing captures → direct threats → quiet moves.

This is not a ranking of move quality. A quiet move may be best. It is a ranking of

how easily branches can be proved.

9.4 Induction over a line

Suppose you claim a three-move combination works. The logic is:

1. At the final leaf, the outcome is winning.
2. On the opponent's preceding turn, every relevant defense reaches a winning leaf.
3. Therefore your preceding move forces a winning continuation.
4. Repeat backward until the first move inherits the proof.

This is why a single unexamined defense breaks the whole argument. A combination is a chain of universal claims about the opponent's replies.

Worked example: calculate the recapture last

You can sacrifice on a defended square, force the king onto a file, and then win the queen with a discovered attack. Many players stop after seeing the discovered attack. Backward induction evaluates the final recapture: after winning the queen, can the opponent take your attacking queen or force perpetual check? Only when the final material and king-safety state is stable does the first sacrifice become proved.

9.5 Known end states as libraries

Endgame knowledge shortens calculation because it labels leaves. If you know that a king-and-pawn position is won, a tactical line ending there can be evaluated immediately. If you know a rook ending is theoretically drawn but practically difficult, you can assign both objective and practical values.

Patterns are not substitutes for calculation; they are compressed terminal evaluations.

Common error

Never end a line on your move. End after the opponent's strongest forcing reply, when all immediate checks, captures, and threats have been resolved.

Exercise 9.1: Terminal target

Choose a tactic. State its terminal claim before listing moves. Then classify every defense by category and explain why the leaf supports the claim.

Exercise 9.2: Threat or force?

Create one position statement that is merely a threat and one that is forced. Explain what the defender can choose in each.

The chapter in one minute

Backward induction starts from a proven leaf and carries the value backward. State the terminal claim, enumerate defensive categories, and stop only at a stable leaf.

Dynamic value: tempo, initiative, and zugzwang

You will treat turns as scarce resources, distinguish development from initiative, understand forcing tempo and loss of options, and model zugzwang as negative action value.

The question underneath the chapter

How can the right to move be an advantage in one position and a burden in another?

10.1 A move spends a turn

Every action has an opportunity cost: while one piece moves, every other possible action is delayed. A move is efficient when it changes several useful relationships at once—developing with tempo, defending while attacking, or improving a piece while preventing a break.

A **tempo** is one unit of move order. Its value is state-dependent. One tempo in a mating race can decide the game. One tempo in a locked position may be used for a waiting move.

10.2 Development, initiative, and force

These terms are related but distinct.

Development.

Mobilizing pieces from low-activity starting squares.

Initiative.

Making threats or creating questions that influence the opponent's choices.

Forcing tempo.

A move that requires a narrow class of replies.

You can be ahead in development but lose the initiative if your pieces do not threaten anything. You can have the initiative with less material if every move creates an urgent problem.

Intuition

Initiative is temporary control over the opponent’s to-do list.

10.3 The initiative ledger

Track who is spending moves on plans and who is spending moves on answers.

	Initiative holder	Responder
Typical move	creates threat, gains space, improves with tempo	parries, retreats, trades attacker
Risk	overextension, insufficient force	passivity, accumulated weakness
Goal	convert before momentum fades	neutralize and seize a free move

The defender often should return material to break the initiative. Material is stored utility; it can be spent to buy time, simplify, or close lines.

10.4 Move-order economics

Two sequences may reach similar positions, but the order changes what the opponent can insert. An **intermediate move** exploits this. Before automatic recapture, ask whether a check, attack on the queen, or stronger capture changes the order.

The same logic governs preparation. If a pawn break will provoke a reply, you may first improve the piece that benefits from that reply. But every preparatory move gives the opponent a turn. Preparation is justified only if its gain exceeds the opponent’s best use of the tempo.

Formal lens

Think of a move’s dynamic contribution as

$$D(a) = \text{useful changes created} - \text{best use of the reply granted.}$$

This is not a numerical engine formula. It is an accounting identity: every non-forcing move purchases benefits while giving the opponent one action.

10.5 Zugzwang: when action has negative value

Normally, having the move is valuable because it expands choice. In zugzwang, every legal move worsens the position. Passing would preserve utility, but passing is not in the action set.

Zugzwang teaches a general lesson: more options are valuable only when at least one option preserves or improves the state. A player may have many legal moves but no useful move.

Worked example: reserve tempi

In a pawn ending, a harmless pawn move can be stored as a reserve tempo. Later, when king opposition matters, the reserve move transfers the obligation to move. The pawn's value is not only its promotion chance; it also functions as a timing instrument.

10.6 Convert or renew

An initiative decays if it does not create durable gains. Ask every two or three moves:

- Can I convert it into material, structure, king exposure, or a favorable ending?
- Can I renew it with another forcing move?
- If neither, should I consolidate before the opponent gains free moves?

Memory hook

A tempo is not "one point." It is the right action at the right moment. Price it by what both players can do with the next turn.

Exercise 10.1: Tempo audit

Take a three-move plan from one of your games. For each preparatory move, state the benefit purchased and the opponent's best use of the granted turn.

Exercise 10.2: Initiative conversion

Find a position where one side has the initiative. Name one way to convert it, one way to renew it, and one defensive method to extinguish it.

The chapter in one minute

Turns are resources. Initiative controls the opponent's to-do list, but it must be converted or renewed. Move order matters because every preparation grants a reply; in zugzwang, the obligation to act becomes a cost.

Part III

Choose Under Risk and Limits

Search under bounded rationality

You will allocate calculation by branch importance, use selective deepening and pruning, recognize horizon effects, and build a repeatable over-the-board search routine.

The question underneath the chapter

If you cannot calculate everything, what deserves your next ten seconds?

11.1 Rationality with a budget

Classical game theory often asks what an ideal optimizer would choose. Human chess asks what a limited optimizer should do with a finite clock, memory, and attention span. This is **bounded rationality**: the decision process itself consumes resources.

The correct goal is not maximum depth everywhere. It is maximum decision quality per unit of attention.

Formal lens

Let $Q(a, d)$ be the estimated quality of candidate a after searching to depth d , and let $C(a, d)$ be its cognitive and time cost. A bounded player seeks a useful balance such as

$$\max_{a,d} Q(a, d) - \lambda C(a, d),$$

where λ rises as clock time becomes scarce. The expression is conceptual: under time pressure, an extra ply of analysis must justify its cost.

11.2 Search where value can flip

A branch deserves more attention when one of four conditions holds:

1. **Forcingness**: checks, captures, and threats narrow replies.

2. **Volatility:** one move can change the result or evaluation sharply.
3. **Irreversibility:** pawn breaks, sacrifices, exchanges, and king moves remove options.
4. **Uncertainty:** your evaluation depends on a fact you have not verified.

A quiet rook improvement in a stable position may need little depth. A pawn sacrifice opening files near your king needs much more.

11.3 Selective deepening

Use a funnel rather than a flood.

Stage 1: Scan. Reconstruct the state and list forcing moves for both sides.

Stage 2: Generate. Create three to five candidates with distinct functions.

Stage 3: Triage. Reject only candidates with clear forcing defects.

Stage 4: Deepen. Calculate the two most promising candidates, following strongest replies.

Stage 5: Stabilize. Continue forcing lines until a quiet leaf.

Stage 6: Compare. Use the same evaluation vector at each leaf.

Stage 7: Blunder check. Run SAFE on the chosen move.

This routine is iterative. If deepening reveals a new tactical fact, return to generation. Do not defend an early favorite merely because you invested time in it.

11.4 Pruning without lying

Pruning saves time by cutting branches unlikely to affect the choice. Good pruning uses evidence:

- a candidate is refuted by a forcing line;
- one move dominates another over the relevant reply set;
- a branch transposes to a position already evaluated;
- a reply is clearly inferior and cannot create a tactical exception;
- the current candidate already fails the required security level.

Bad pruning uses emotion or habit: “they would never find that,” “this move is ugly,” or “I always play this structure.”

Favorite-line bias

The more attractive your idea feels, the more deliberately you should search for a refutation. Investment in a line is not evidence that the line works.

11.5 Horizon and quiescence

A **horizon effect** occurs when analysis stops just before a consequence appears. You “win” a pawn at the leaf, but one move later your queen is trapped. A position is **quiescent** when immediate forcing actions have settled enough for static evaluation to be meaningful.

Before stopping, ask:

- Are there unresolved checks or forcing captures?
- Is a piece en prise or trapped?
- Is a promotion race still concrete?
- Is the material gain temporary?
- Can the opponent force a structural change immediately?

11.6 Chunking and pattern libraries

Experts do not merely calculate faster; they recognize meaningful chunks. “Minority attack,” “back-rank weakness,” and “outside passed pawn” compress many relationships into one retrieval cue. A pattern is valuable when it suggests candidates and evaluation questions, not when it replaces verification.

Build patterns from your own games by storing three items:

1. the trigger configuration;
2. the typical action;
3. the exception that would make the action fail.

Worked example: attention allocation

You have six minutes for twelve moves. The position is closed, with no checks or captures. A pawn break would open your king; a rook doubling move is

reversible. Spend most calculation on the pawn break because it is volatile and irreversible. Verify the rook move tactically, but do not search equal depth merely for symmetry.

Memory hook

Calculate by danger and irreversibility, not by the number of legal moves.

Exercise 11.1: Search budget

For a position from your games, rank three candidates by forcingness, volatility, irreversibility, and uncertainty. Allocate a hypothetical two-minute budget among them and explain why.

Exercise 11.2: Horizon repair

Find a line where your original analysis stopped too early. Extend it until a quiet leaf and state which unresolved forcing feature caused the error.

The chapter in one minute

Bounded rationality means attention is part of the problem. Search volatile, forcing, irreversible, and uncertain branches more deeply. Prune with evidence, and stop only at a quiescent leaf.

Expected utility and practical chances

You will combine objective security with estimates of human response, distinguish probability from value, and compare moves by expected score without disguising tactical refutations.

The question underneath the chapter

When should you choose the move a human is least likely to handle rather than the move an engine ranks first?

12.1 From worst case to weighted branches

Minimax assumes the opponent selects the reply worst for you. Expected utility weights outcomes by their probability:

$$EU(a) = \sum_i P(r_i \mid a) U(z_i),$$

where r_i is an opponent reply and z_i the resulting outcome or evaluated leaf.

This lens matters because humans are fallible. A move can create many difficult decisions, and a theoretically equal defense may be hard to find. But expected utility must be used only after tactical cliffs are identified.

Intuition

Minimax asks, “Can this fail?” Expected utility asks, “How often, and what happens in the other branches?”

12.2 Probability is conditional

Do not assign a probability to “they blunder.” Estimate a particular reply under particular conditions:

- Is the reply forcing or quiet?
- Is it natural by general principles?
- Does it require seeing a hidden intermediate move?
- How much clock time remains?
- Does it match the opponent's style and preparation?
- Are there many plausible alternatives?

A one-move check is likely to be found. A counterintuitive retreat that permits a temporary sacrifice may be much less findable.

12.3 Objective floor, practical ceiling

Use a two-stage test.

Stage 1: objective floor. What is the candidate's value against the strongest reply you can find? If that floor is an unacceptable forced loss, reject the move unless event utility truly requires gambling.

Stage 2: practical ceiling. Among candidates with acceptable floors, compare how difficult the opponent's decisions are, how easy your own continuation is, and how the clock shapes errors.

This prevents "trap chess," where a bad move is justified solely because the opponent might miss the refutation.

12.4 Error asymmetry

A position has favorable **error asymmetry** when your moves are natural and numerous while the opponent must repeatedly choose precise moves. It has unfavorable error asymmetry when you have only-moves and the opponent can improve almost automatically.

Compare two equal positions:

	Position A	Position B
Your task	find one of several improving moves	find repeated defensive only-moves
Opponent’s task	solve concrete threats	improve pieces naturally
Clock effect	helps your pressure	magnifies your danger
Practical utility	often high	often low

12.5 A simple decision table

Suppose a move leads to three broad reply classes. Use rough categories rather than fake precision.

Reply class	Likelihood	Leaf utility	Contribution
Best defense	low–medium	equal	modest
Natural defense	medium	pleasant edge	positive
Error	low	winning	positive but discounted

Then stress-test the estimate. If the “best defense” is actually easy, raise its probability. If the winning branch still requires difficult technique, lower its utility.

12.6 Risk dominance in practical chess

When two moves have similar expected utility, prefer the one that remains acceptable across a broader range of probability estimates. If Move A is best only when your guess that the opponent misses a defense is 70 percent, while Move B is good from 20 to 80 percent, B is probability-robust.

Worked example: the hidden defense

You can enter a tactical complication where one quiet retreat equalizes. Your opponent has two minutes, and the retreat contradicts the natural impulse to recapture. The line may have practical value. First verify that the equalizing retreat does not leave you worse. Then compare the line with a simpler move that gives a small stable edge. Your event need and personal calculation confidence determine the final utility ranking.

Common error

Never replace “I did not calculate the defense” with “the defense is unlikely.” Findability is a second question, not a substitute for analysis.

Exercise 12.1: Expected-score sketch

For two candidate moves, create three reply classes, assign rough probabilities that sum to one, and assign utilities. Compute the expected utility, then change one probability to see whether the decision is robust.

Exercise 12.2: Error asymmetry

Choose a position that is objectively equal. List the number and naturalness of acceptable moves for each side. Which side has the easier practical task?

The chapter in one minute

Expected utility weights branches by practical likelihood, but only after you establish an acceptable objective floor. Prefer favorable error asymmetry and decisions robust to uncertain probability estimates.

Mixed strategies and the value of unpredictability

You will understand why randomization is unnecessary for perfect play in a single chess game yet useful across repeated encounters, opening preparation, and opponent exploitation.

The question underneath the chapter

If one move is best, why should a player ever vary?

13.1 Pure and mixed strategies

A **pure strategy** chooses a definite action at each decision. A **mixed strategy** assigns probabilities to alternatives. In a finite deterministic perfect-information game, optimal play does not require randomization merely to resolve hidden simultaneous choice. If one continuation guarantees the best game-theoretic value, you may choose it.

Human competition adds repeated interaction and limited preparation. If your opponent knows you always answer 1.e4 with the same narrow system, they can invest all preparation there. Varying can reduce the return on their preparation.

13.2 Unpredictability is not randomness for its own sake

A useful mix contains moves that are individually sound. Randomly selecting an inferior move merely to surprise is usually expensive. The goal is to make exploitation difficult while preserving utility.

Imagine two opening systems:

	System A	System B
Objective quality	sound	sound
Preparation cost	already learned	moderate
Style	forcing	strategic
Opponent prep	likely targeted	less expected
Role in mix	main line	diversification

A 70–30 mix may force the opponent to divide study time. The exact probabilities need not be generated by dice; they can be planned frequencies across a season.

13.3 The exploitability test

Ask:

If my next opponent knew my full choice rule, could they improve their preparation or game plan against me?

If yes, you are exploitable. Exploitability can arise from openings, draw-offer habits, time usage, recurring recapture choices, or a tendency to avoid queen trades.

13.4 Mixing at different levels

Repertoire level. Maintain two sound answers to a major opening branch.

Move-order level. Reach the same structure through different sequences to avoid prepared forcing lines.

Position-type level. Be capable of both tactical and strategic play so opponents cannot steer you cheaply.

Clock level. Varying thought time can avoid signaling that a move surprised you, though clock preservation is usually more important than theatrical deception.

13.5 When not to mix

Do not mix when:

- one move is clearly and safely superior;

- your secondary option is underprepared;
- the position is tactical and only one move survives;
- variety would dilute scarce training time below competence;
- event utility strongly favors one position type.

Consistency is valuable when it builds deep pattern knowledge. The strategic question is whether the learning benefit of specialization exceeds the exploitability cost.

Formal lens

In a repeated preparation game, let p be the frequency with which you choose System A. Your opponent allocates preparation to maximize expected gain against that frequency. A useful mix makes the opponent roughly indifferent between concentrating on A and covering B; otherwise they can profitably re-allocate effort.

Worked example: surprise with substance

You normally play a sharp main line. For a match, you add a quieter move that reaches a structure you understand. The surprise is not valuable because it is rare. It is valuable because it forces the opponent to solve fresh but sound problems while you retain familiar plans.

Memory hook

The best surprise is a good move whose preparation cost falls more on the opponent than on you.

Exercise 13.1: Exploitability audit

List three predictable habits in your chess. For each, state how an informed opponent could exploit it and whether to improve, conceal, or deliberately keep the habit.

Exercise 13.2: Repertoire mix

Choose one major opening decision. Design a two-option mix in which both choices are sound, serve different practical roles, and fit within your study budget.

The chapter in one minute

Perfect play in one chess game needs no randomization, but repeated human competition rewards sound variety. Mix to reduce exploitability, not to excuse inferior moves.

Time controls and clock utility

You will treat clock time as a state variable, price thinking time by decision importance, and use time thresholds, reserve rules, and low-time protocols.

The question underneath the chapter

How much is one minute worth in the current position?

14.1 The clock changes the game

Time is not merely a constraint outside the board. It changes the probability of future errors, the feasibility of complex plans, and the value of simplification. A complete practical state is (s, t_W, t_B) : board state plus both clocks.

One minute has variable marginal value. In a forced sequence, extra thought may add little. Before an irreversible sacrifice, it may be decisive. Near move forty in a classical time control, preserving enough time to reach the increment can dominate a small positional gain.

14.2 Think in decisions, not moves

Equal time per move is inefficient. Allocate time by decision importance:

Decision type	Spend more time	Spend less time
Forcing tactic	many branches or unclear leaf	forced recapture, verified sequence
Pawn break	irreversible, opens kings	standard break with known response
Exchange	changes ending/result prospects	clearly favorable forced trade
Quiet move	several plans compete	one natural safe improvement
Defense	only-move risk, checks	simple neutralization

14.3 The time budget equation

A rough budget is

average available = $\frac{\text{time remaining} - \text{reserve}}{\text{moves to next control or estimated moves}}$.

Do not mechanically spend that amount. Use it as a baseline, then borrow for critical decisions and repay on forced moves.

Set a **reserve**: time you refuse to spend before a future phase. For example, preserve five minutes for the final ten moves before a control, or preserve enough time to execute a technical ending without panic.

14.4 Stop rules

Overthinking often occurs because the player has no criterion for ending analysis. Stop when:

- one candidate has an acceptable security level and competitors are refuted;
- forcing lines reach stable leaves;
- further analysis repeats the same branches without new information;
- the marginal value of thought is lower than the future value of the time;
- your process is deteriorating and a robust move is available.

A stop rule prevents analysis from becoming anxiety management.

14.5 Low-time protocol

When time is scarce, shrink the process but do not abandon it:

1. Reconstruct checks against both kings.
2. Identify all loose pieces and immediate captures.
3. Choose one forcing and one robust candidate.
4. Reject any move that fails SAFE.
5. Prefer simple execution and favorable error asymmetry.

With increment, make a legal safe move before the clock reaches the point where physical execution becomes risky. Do not spend an increment repeatedly proving a move you already know you will play.

14.6 Clock as signal and weapon

Thinking time can reveal surprise or uncertainty, but trying to fake signals is usually less valuable than preserving time. More important is how your move changes the opponent's time problem. A move with several plausible replies may force them to spend; a forcing move may make their decision easy. Complexity is not automatically pressure if the opponent's best response is obvious.

Worked example: buying future accuracy

You have eight minutes for eight moves. A stable move preserves a slight edge; a sharp move may win more but requires three future only-moves. Even if both are objectively sound, the stable move can have higher utility because it converts current time into future accuracy. You are not “wasting an advantage”; you are pricing execution risk.

Common error

Do not use almost all your time to reach a winning position you can no longer execute. Utility belongs to the final result, not the evaluation at move thirty.

Exercise 14.1: Time map

Take one of your games and mark three critical decisions. Compare actual time spent with forcingness, volatility, irreversibility, and uncertainty. Where was time misallocated?

Exercise 14.2: Design a protocol

Write a personalized low-time routine of no more than five steps and a reserve rule for your usual time control.

The chapter in one minute

Clock time is part of the state. Spend by decision importance, maintain a reserve, use stop rules, and under pressure shrink the process to checks, loose pieces, robust candidates, and SAFE.

Part IV

Think About Minds

Beliefs and Bayesian updating

You will distinguish board evaluation from beliefs about an opponent, update those beliefs from evidence, avoid stereotypes, and use opponent models only when they change a close decision.

The question underneath the chapter

What did the opponent's last move reveal beyond the squares it changed?

15.1 The board state and the player type

The board is visible. The opponent's internal state is not. You may be uncertain about their opening knowledge, tactical alertness, risk attitude, fatigue, time-management habits, and preferred structures. Game theory represents such hidden characteristics as a **type**.

Let θ denote an opponent type, such as “well prepared in this forcing line” or “unfamiliar with this pawn structure.” You begin with a prior belief $P(\theta)$. Each move supplies evidence e , producing a posterior belief:

$$P(\theta | e) = \frac{P(e | \theta)P(\theta)}{P(e)}.$$

You do not need numerical calculation at the board. The formula teaches direction: evidence matters in proportion to how differently likely it is under competing explanations.

15.2 Strong and weak evidence

A rapid sequence of accurate theoretical moves is evidence of preparation because it is more likely from a prepared player than an unprepared one. One aggressive pawn push is weak evidence of an “attacking personality” because many board reasons could explain it.

Observation	Possible update	Alternative explanation
Instant accurate novelty response	preparation more likely	forced or obvious move
Long think in known structure	unfamiliarity more likely	recalling a deep line carefully
Repeated queen-trade offers	draw preference more likely	trades objectively best
Fast risky moves in time trouble	risk tolerance may be high	desperation from the position
Quiet defensive retreat	resourcefulness more likely	only legal defense

Good updating keeps alternative explanations alive.

15.3 Priors should be broad, not prejudiced

Ratings, past games, and known repertoire can create useful priors. They must not become rigid stories. A lower-rated player can calculate a position accurately; an attacking player can choose a quiet defense. Board evidence dominates labels.

Use three confidence levels:

Low confidence.

One ambiguous move; do not change strategy.

Medium confidence.

Several consistent observations; use as a tie-breaker.

High confidence.

Direct preparation evidence or repeated behavior; allow meaningful adjustment, still with objective safety.

15.4 Bayesian candidate choice

Opponent beliefs matter most when objective candidates are close. Suppose two sound moves lead to different tasks:

- Move A enters a forcing theoretical line.
- Move B enters a strategic structure with many plans.

If the opponent has played the first fifteen moves instantly and accurately, increase the probability they are prepared. B may gain practical utility. If B is objectively inferior, the belief should not magically repair it; the update changes a close comparison, not the rules of chess.

15.5 Learning from time usage

Time is evidence, but noisy evidence. A long think can mean surprise, calculation, indecision, or an attempt to remember. A fast move can mean preparation, obviousness, or carelessness. Combine time with move quality and structure familiarity.

A useful question is:

Which hypothesis makes the observed move-and-time pair least surprising?

Worked example: updating without overreacting

Your opponent plays an unusual opening move instantly. Prior: perhaps it is preparation. Evidence after two more moves: they spend a long time on a standard thematic break and choose an inaccurate plan. Posterior: the first move may have been a memorized surprise with shallow follow-up. You now prefer a normal sound continuation that asks structural questions rather than trying to “refute” the opening immediately.

Narrative overfitting

One move is rarely a personality test. Update beliefs gradually and keep the board’s objective demands primary.

15.6 The belief ledger

Between games, not during every move, track:

1. hypothesis about the opponent;
2. evidence for and against;
3. confidence level;
4. decision that would change if the hypothesis were true;
5. post-game check against reality.

This turns opponent modeling into calibrated learning rather than folklore.

Exercise 15.1: Update a belief

Choose one observation from a game. State a prior hypothesis, the likelihood of the observation under that hypothesis and an alternative, and the direction of your update.

Exercise 15.2: Tie-breaker only

Find two objectively similar candidates. Explain how one plausible opponent belief would change their practical ranking and why it should not override a tactical refutation.

The chapter in one minute

A player type is hidden; the board is not. Update beliefs when evidence is more likely under one explanation than another, keep alternatives alive, and use beliefs mainly to break close objective ties.

Signals, threats, and credible commitment

You will understand what moves can signal, distinguish an objective threat from a psychological message, test credibility through cost and commitment, and avoid empty “intimidation” chess.

The question underneath the chapter

When does a move communicate a plan the opponent should rationally believe?

16.1 Moves do two jobs

Every move changes the board. It can also communicate information about intentions, preparation, and willingness to enter certain positions. In game theory, a **signal** is an observed action that may convey hidden information about type or intention.

Chess signals are constrained by perfect information: a move cannot make a false attack become objectively real. If a sacrifice does not work, confidence does not repair it. Yet a move may still alter beliefs and decision difficulty.

16.2 Cheap talk and costly signals

Cheap talk is communication that does not directly change payoffs or constrain future action. Facial confidence, fast play, or a dramatic gesture may influence a human but carries little strategic credibility.

A **costly signal** commits real board resources. Advancing a kingside pawn reveals willingness to weaken squares for space or attack. Sacrificing material proves commitment because the material cannot simply be recalled. A queen retreat into a defensive role may credibly signal that the player values consolidation.

The signal is credible when faking it would be too costly for a type without the

claimed plan.

Formal lens

A signal separates types when different types prefer different signals because their costs differ. In chess terms, a well-prepared attacker may accept a pawn sacrifice whose compensation they understand, while an unprepared player pays a larger practical cost for the same commitment.

16.3 Threats as incentive design

A threat is not merely an announced future move. It changes the opponent's incentives by attaching a cost to inaction. A good threat often improves your position even if answered:

- it forces a passive defense;
- it provokes a weakening pawn move;
- it gains a useful tempo;
- it distracts a defender;
- it induces an unfavorable exchange;
- it restricts the opponent's candidate set.

The best threats are **dual-purpose**: the threatened action is real, and the move also improves a piece or structure.

16.4 Credibility test

Before relying on a threat, ask:

1. If ignored, does the threatened move actually improve my utility?
2. Can the opponent answer while advancing their own plan?
3. Am I willing and able to carry out the threat after the board changes?
4. Does my threat depend on the opponent choosing a convenient defense?
5. What did I give up to create it?

A threat that you should not execute is not credible. The opponent may rationally ignore it.

16.5 Commitment devices

A commitment device removes your future freedom in order to influence current behavior. Castling to opposite wings, fixing the center, or sacrificing a pawn can commit the game to a type of struggle. Commitment can be valuable because it makes a plan credible and coordinates your pieces. It can be dangerous because new information cannot be accommodated.

Preserve flexibility when uncertainty is high. Commit when the gain from coordination or force exceeds the option value lost.

Worked example: the pawn storm signal

You play h4 near the enemy king. Board effect: gain space and prepare h5. Signal: willingness to attack. Credibility depends on the center. If the opponent can open the center against your uncastled king, h4 may be an empty signal and a real weakness. If the center is locked and your pieces can join, the commitment is coherent. The board validates the message.

16.6 Inducing choices

Sometimes the goal is not to force one reply but to offer a menu in which every reply concedes something. A pin may let the opponent choose between structural damage and loss of activity. A passed pawn may force a rook into passivity, allowing invasion elsewhere.

This is chess as mechanism design in miniature: shape the opponent's action set so their self-interested best response advances your plan.

Memory hook

A credible threat is a future action you would genuinely want to execute if ignored.

Exercise 16.1: Signal or board effect?

Choose an aggressive move. Separate its objective board effects from the beliefs it might create. Which effects remain if the opponent feels no fear?

Exercise 16.2: Credibility audit

Find a threat from one of your games. Apply the five credibility questions. Was the threat forcing, useful when answered, or empty?

The chapter in one minute

Moves can signal, but the board determines whether the signal has substance. Credibility comes from a real beneficial follow-through, often supported by cost or commitment.

Prophylaxis and common knowledge

You will use recursive opponent modeling, distinguish prevention from passivity, and choose between stopping, permitting, provoking, and exploiting an opponent's plan.

The question underneath the chapter

What changes when both players know what the other player wants?

17.1 Thinking one mind deeper

Prophylaxis begins with the opponent's desired action. But strong chess goes further:

1. What do they want?
2. What do they think I want?
3. How will they prevent my plan?
4. Can I exploit the preventive move they are likely to choose?

This recursive reasoning resembles higher-order beliefs in game theory. "I know" is first order. "I know that they know" is second order. Chess does not require infinite recursion; usually two levels reveal the key interaction.

17.2 Common knowledge

A fact is common knowledge when both players know it, both know that both know it, and so on. An obvious weak back rank can be common knowledge. The strategic battle then moves from discovering the weakness to exploiting the incentives it

creates: perhaps neither side can move a rook because both understand the tactical consequence.

Common knowledge can create stability. A tactical resource that is visible to both sides may never be played, yet it constrains moves for many turns.

17.3 Four responses to an opponent's plan

Do not automatically prevent everything. Choose among:

Stop.

Make the plan impossible when it would be strong and prevention is cheap.

Permit.

Allow it when the resulting position is harmless or favorable.

Provoke.

Encourage it because the plan creates a weakness or overcommitment.

Race.

Ignore it because your own action is faster or more valuable.

Prophylaxis is therefore comparative, not fearful. The question is the utility of their plan relative to the cost of your response.

17.4 The free-move experiment

At each quiet decision, imagine the opponent can move twice. What would the first move prepare? This reveals latent plans that are not yet threats.

Then ask a second question: if you stop that plan, what new freedom do they gain? A move that prevents one break may abandon another square. Good prophylaxis alters incentives without creating a larger concession.

Worked example: invite the break

The opponent wants a pawn break that seems to attack your center. You can prevent it with a weakening pawn move. Analysis shows the break actually opens a file for your rook and leaves their pawn isolated. The correct prophylactic choice is to permit or even provoke the break. Understanding the opponent's goal does not imply obeying it.

17.5 Useful restriction

Restriction reduces the opponent’s effective action set. A square can be legally available but strategically unusable because entering it loses material. Restriction works through:

- controlling breaks;
- taking away piece squares;
- fixing targets;
- tying defenders;
- preserving a tactical punishment;
- exchanging the opponent’s active piece.

The ideal restrictive move also improves your own position. Purely defensive prevention may cede the initiative.

17.6 Mutual plans as a matrix

For a difficult position, compare plans rather than single moves:

	They execute plan	They prevent yours
You execute yours	race: compare speed and force	test whether your plan still works
You prevent theirs	compare cost of prevention	maneuvering equilibrium

This matrix prevents vague thinking. Name the plans, the move orders, and the resulting concessions.

Common error

Do not spend a useful move stopping an opponent’s bad plan. Prevention has an opportunity cost.

Exercise 17.1: Four responses

Identify an opponent plan from a game. Analyze one move each that would stop, permit, provoke, or race it. Rank the four responses.

Exercise 17.2: Second-order belief

State what your opponent wanted, what they likely believed you wanted, and how that belief shaped their move. What alternative could you have prepared?

The chapter in one minute

Prophylaxis is not automatic prevention. Model the opponent's goal, then choose whether to stop, permit, provoke, or race it. Think one level deeper about what they believe you want.

Biases, emotions, and distorted utility

You will recognize common decision biases in chess, separate emotion from information, and use process guardrails to recover after surprise or error.

The question underneath the chapter

Which opponent is more dangerous: the player across the board or the story in your head?

18.1 Bounded rationality is systematic

Human limitations are not random. We make predictable errors. Recognizing them creates a practical advantage because a guardrail can be designed before emotion arrives.

18.2 Eight recurring biases

Confirmation bias. Search for moves that prove your idea and ignore refutations.

Guardrail: after finding a move, spend one pass trying to defeat it.

Anchoring. Stay attached to an earlier evaluation such as “I am winning.”

Guardrail: rebuild the vector after every tactical or structural change.

Sunk-cost bias. Continue an attack because you already sacrificed material or time.

Guardrail: evaluate from the current state; previous investment is gone.

Loss aversion. Fear giving back material even when returning it secures the king or a draw. *Guardrail:* compare terminal utility, not the pain of the concession.

Recency bias. Overweight the opponent's last tactic or your last blunder. *Guardrail:* ask what the full position, not the last event, demands.

Availability bias. Choose a move because a memorable pattern resembles the position. *Guardrail:* state the pattern's required conditions and verify them.

Status bias. Alter objective judgment because of rating, title, or reputation. *Guardrail:* write the board evaluation before opponent-type adjustments.

Action bias. Feel compelled to attack or change the structure when waiting is best. *Guardrail:* include one improving or waiting candidate.

18.3 Emotion as data, not command

Fear may signal that you have not checked a forcing line. Excitement may signal a tactical pattern. These feelings are useful prompts to inspect the state, but poor decision rules.

Use the sequence:

Name the feeling → **Locate** the board source → **Test** it with replies → **Choose** by utility.

If no board source appears, the emotion may be residue from rating, previous mistakes, or imagined expectations.

18.4 Tilt as preference corruption

After a blunder, a player may prefer a low-probability swindle to a high-probability draw because accepting the draw feels like admitting failure. The preference ordering has shifted from "maximize score" to "repair identity." That shift is rarely deliberate.

A reset protocol:

1. Sit physically still and take one slow breath.
2. Say the material count and immediate threats.
3. Delete the previous evaluation.
4. Identify the new result goal.
5. Choose one robust candidate before seeking tricks.

18.5 The narrative audit

Players often tell stories: “They are attacking,” “I have the bishop pair,” “This pawn must be weak.” Convert every story into falsifiable board claims.

Story	Testable claim
They are attacking	list attackers, entry squares, forcing moves, and defenders
My bishop pair is strong	show open diagonals and targets on both wings
Their pawn is weak	show that it can be fixed, attacked, and not advanced
I have compensation	identify time, king safety, activity, or structure equal to material

Worked example: fear of the higher-rated opponent

You see a sound simplifying move but reject it because the opponent “must know the ending better.” This may be true, but it is not yet decision-relevant. Evaluate the ending. Compare it with the tactical risk of alternatives. Use rating only as a small belief adjustment after objective work, not as a replacement for it.

Memory hook

Feelings can choose the question. They should not choose the move.

Exercise 18.1: Bias diagnosis

Select one serious error and classify the dominant bias. Write a one-sentence guardrail that would have interrupted it.

Exercise 18.2: Rewrite a story

Take three narrative statements from your annotations and convert each into measurable board claims.

The chapter in one minute

Biases systematically distort search and utility. Name emotion, locate its board source, test it, and choose by current future outcomes. Use prewritten guardrails after surprise and error.

Repeated encounters, reputation, and adaptation

You will view matches and tournaments as repeated games, understand reputation as a strategic state, adapt without overfitting, and manage information across encounters.

The question underneath the chapter

How can today's opening choice change the opponent's preparation next month?

19.1 One game changes the next

Formal over-the-board chess ends at the result. Competitive relationships continue. Opponents remember openings, time habits, risk choices, draw behavior, and emotional reactions. A match or tournament series is a repeated game with learning.

Your broader state includes a public history:

$$H = (\text{past openings, results, visible habits, reputation}).$$

This history changes beliefs and preparation incentives.

19.2 Reputation as a commitment asset

A reputation for tactical alertness can deter speculative sacrifices. A reputation for accepting early draws may encourage opponents to offer them. A reputation for one narrow opening can invite deep preparation.

Reputation is useful when it credibly reflects skill or policy. It is costly when it makes you predictable. The solution is not image management at the expense of chess, but awareness that repeated behavior becomes information.

19.3 Exploit, explore, or conceal

Across repeated games you face three learning choices:

Exploit.

Use a known opponent weakness now.

Explore.

Choose a line that reveals how they respond, accepting some immediate uncertainty.

Conceal.

Avoid revealing a prepared weapon before a more important game.

In most ordinary tournaments, maximizing the current game's score dominates elaborate concealment. In a match, information management can matter more because the same branches may recur.

19.4 Adaptation loops

After each encounter:

1. update your beliefs about the opponent;
2. predict how they will update beliefs about you;
3. identify what preparation each side now has an incentive to change;
4. choose whether to repeat, vary, or reverse course.

A common match tactic is a second-level adaptation: repeat a line long enough that the opponent prepares an improvement, then switch to a different system. This works only if the new system is sound and prepared; otherwise strategic theater replaces chess quality.

19.5 Do not overfit one game

A single result contains noise from tactics, time trouble, and mood. Distinguish:

- **stable traits:** recurring time usage, long-term repertoire, structural preferences;
- **state effects:** fatigue, event situation, surprise, one tactical oversight;
- **position effects:** the board simply demanded the observed move.

Update more after repeated evidence or a highly diagnostic choice.

19.6 Tournament ecology

Even when players do not meet repeatedly, standings create indirect strategic interaction. Your result changes pairings, tie-breaks, and what other players need. Draw incentives may shift by round. The correct unit of utility is sometimes the event, not the isolated game.

Still, never assume collusion or behavior outside the rules. Model only legitimate incentives and choose within fair-play standards.

Worked example: repeat the opening?

You won a game because the opponent mishandled a structure, not because the opening was objectively strong. In the rematch, repeating may invite targeted repair. Ask whether the line remains sound after their likely improvement. If yes, repetition can test whether their understanding changed. If no, the previous result is a poor reason to repeat.

Common error

Results are evidence about decisions, not proof that every decision was good. A bad opening can win; a good plan can lose after a later blunder.

Exercise 19.1: Second-game plan

Choose a past opponent you may face again. List what they learned about you, what you learned about them, and one likely adaptation by each side.

Exercise 19.2: Trait or state?

Classify five observations from an opponent's games as stable trait, temporary state, or position effect. State your confidence.

The chapter in one minute

Repeated play creates a public history that shapes preparation and beliefs. Exploit stable evidence, explore when learning is valuable, conceal only when stakes justify it, and do not overfit one result.

Part V

Apply the Framework

The opening as a prior, not a script

You will treat opening knowledge as a prior over promising moves, understand development as option creation, use move-order logic, and know when to leave memory and start thinking.

The question underneath the chapter

What is opening knowledge for if the opponent can deviate at any move?

20.1 A prior narrows search

An opening repertoire does not solve chess. It gives a prior belief about which moves are likely sound and which structures are likely to arise. Before calculation, a familiar move may receive more attention because accumulated theory and experience support it. New board evidence must still update the prior.

The mistake is to treat memory as a command after the position has changed. A remembered move belongs to a particular state, including move order. If one piece stands on a different square, the tactical justification may disappear.

Intuition

Opening memory proposes. The current position disposes.

20.2 Opening goals as option creation

The familiar principles can be expressed game-theoretically.

Control the center.

Increase useful actions and restrict the opponent's piece routes.

Develop pieces.

Expand the effective action set and coordinate future strategies.

Secure the king.

Reduce the opponent's forcing options.

Avoid premature queen moves.

Do not offer the opponent development with tempo.

Connect rooks.

Improve coordination and enable contested-file strategies.

Development is valuable not because a knight "must" move by move three, but because active pieces create options and reduce tactical vulnerability.

20.3 Move orders as information games

Two sequences may reach the same structure while revealing different information or allowing different deviations. A flexible move can ask the opponent to commit first. A forcing move can prevent them from choosing a favorable setup.

Use three move-order questions:

1. What extra option does this order permit the opponent?
2. What information about my structure or piece placement am I revealing?
3. If the moves transpose, who gained or lost a useful tempo?

20.4 Preparation as a contingent strategy

Do not memorize one line. Store a decision tree:

- opponent's main strategic choices;
- your response and its purpose;
- tactical trigger positions;
- typical pawn breaks;
- bad-piece routes;
- endgames likely after exchanges;
- the first point where independent thought is required.

For every remembered move, attach a reason and a refutation condition: "This move works because the center is closed; if the e-file is open, recheck king safety."

20.5 Leaving book well

You are out of preparation when one of these occurs:

- you cannot explain the remembered move;
- the opponent’s move changes a key tactical condition;
- you remember two lines but not which state belongs to which;
- clock behavior suggests the opponent knows the forcing branch deeply;
- a practical alternative fits your utility better.

At that moment, reset: MODEL the state, generate CCT+I, and search replies. Do not spend ten minutes trying to recover a memory that may be false.

Worked example: the poisoned memory

You remember a thematic pawn sacrifice from a similar opening. In the known line, the opponent’s knight blocks the queen’s defense. In the current move order, the knight has not moved and the queen protects the capture square. The pattern is a prior, but the state update refutes the memory. Choose a normal developing move.

20.6 Repertoire economics

Study time is scarce. A good repertoire balances:

Dimension	Narrow repertoire	Broad repertoire
Pattern depth	high	distributed
Preparation burden	concentrated	large
Predictability	higher	lower
Adaptability	lower	higher
Best for	building foundations	repeated high-level opposition

Most improving players benefit from a stable core plus one sound alternative. Depth creates usable patterns; limited variety reduces exploitation.

Memory hook

Know the reason, the structure, and the exit point from memory. A move without its conditions is not knowledge.

Exercise 20.1: Turn a line into a tree

Choose one opening line you know. Rewrite it as three opponent choices, your response to each, the purpose of each response, and one condition that would invalidate the move.

Exercise 20.2: Repertoire portfolio

Evaluate your repertoire for pattern depth, study cost, predictability, and practical fit. Name one core system and one diversification option.

The chapter in one minute

Openings are priors and contingent strategies, not scripts. Development creates options; move order controls information and deviations. Leave memory as soon as you cannot explain the move's conditions.

The middlegame as a portfolio of imbalances

You will identify the position's dominant imbalance, allocate pieces to roles, compare plans as portfolios of gains and risks, and use switches when the opponent overdefends.

The question underneath the chapter

How do you choose a plan when no tactic forces the answer?

21.1 An imbalance is a persistent asymmetry

A middlegame plan grows from differences: one side has more space, a safer king, a better minor piece, a weak pawn target, a file, a passed pawn, or an initiative. If every feature were symmetric, there would be little reason to prefer one wing or exchange.

First name the **dominant imbalance**: the feature most likely to determine future utility. Then identify supporting imbalances and liabilities.

Imbalance	Natural plan	Opponent's counter-incentive
Space advantage	restrict, improve, switch wings	seek pawn break and exchanges
Better minor piece	preserve, create targets on its color	trade it or change structure
Weak enemy pawn	fix, attack, add second target	advance, exchange, seek activity
Safer king	open lines, add attackers	trade queens or counter in center
Passed pawn	support and distract	blockade and attack base
Initiative	force and convert	return material, simplify, countercheck

21.2 Plan portfolios

A plan is a portfolio of expected gains, risks, time costs, and options. Compare plans using five columns:

1. target created or attacked;
2. worst piece improved;
3. opponent counterplay permitted;
4. irreversible commitments required;
5. favorable ending or tactical transition available.

A plan that attacks one weakness but activates all the opponent's pieces may have poor net value. A quieter plan that fixes the weakness and improves your worst piece can dominate over time.

21.3 Role assignment

Pieces are resources that need jobs. For each piece, state its current and desired role:

Attack.

Create direct threats against king or targets.

Restrict.

Deny squares, breaks, or routes.

Defend.

Protect a critical piece, square, or king zone.

Transform.

Support a pawn break or exchange that changes the structure.

Reserve.

Remain flexible to switch wings or answer a break.

The “worst piece” is often the one with no meaningful role, not merely the one with few legal moves.

21.4 Create a second weakness

If the opponent can defend one fixed target with all pieces, your attack may stall. Switching to a second wing changes their allocation problem. The defender cannot be locally strong everywhere.

The switch works only after the first weakness has tied resources. Attacking everywhere at once without fixing anything merely disperses your own pieces.

Intuition

A weakness matters not only because it can fall, but because it taxes the opponent's pieces.

21.5 Pawn breaks as state transitions

A pawn break changes the game's geometry: files open, diagonals clear, squares weaken, and passed pawns appear. Treat it as a high-volatility state transition.

Before a break, calculate:

- capture, push, and ignore responses;
- which files and diagonals open;
- which piece improves most for each side;
- whether the center opens against a king;
- the resulting pawn islands and passed pawns;
- move-order insertions.

Worked example: attack the base or improve the attacker?

You can add a rook to a backward pawn, but the rook would be tied to defending your own back rank. An alternative king move solves the back rank and prepares rook doubling. The immediate attack looks active; the preparatory move improves the portfolio by covering a liability and preserving the same plan. The opponent's best response during the extra tempo decides whether preparation is justified.

21.6 Plan failure and switching

A plan is a hypothesis. Update it when:

- the opponent neutralizes the dominant imbalance;
- a tactical event changes king safety or material;
- a piece exchange changes which minor piece is superior;
- a pawn break transforms the structure;
- the cost of continuing exceeds the target's value.

Do not continue a plan merely for narrative consistency. Strategy is contingent.

Exercise 21.1: Imbalance map

From a middlegame, list all persistent asymmetries. Select the dominant imbalance and design two plans with different risk and option profiles.

Exercise 21.2: Piece jobs

Assign a current and desired role to every piece on one side. Which piece has no useful job, and what route would give it one?

The chapter in one minute

Middlegame plans grow from persistent asymmetries. Name the dominant imbalance, assign jobs, compare plans as portfolios, and switch when the position changes.

Sacrifice as intertemporal exchange

You will classify compensation, value sacrifices across time, distinguish investment from speculation, and prove a sacrifice against acceptance, refusal, and counterplay.

The question underneath the chapter

How can giving up material increase utility?

22.1 Material now for resources later

A sacrifice exchanges immediate material for future resources: time, king exposure, activity, structure, passed pawns, control, or a forced terminal result. It is **intertemporal** because the costs and benefits arrive at different moments.

A sacrifice is rational when the discounted value of future compensation exceeds the material and options surrendered.

Formal lens

Think schematically:

$$\text{Net sacrifice value} = \sum_{t=1}^H \delta^t C_t - M_0 - R,$$

where M_0 is material given now, C_t is compensation at future time t , δ discounts delayed or uncertain benefits, and R is execution and refutation risk. A forcing mate has little discount. Vague long-term pressure has a large discount.

22.2 Compensation categories

Name compensation precisely:

- forced mate or perpetual check;
- recovery of material by force;
- exposed king and durable attacking access;
- lead in development and initiative;
- trapped or restricted enemy piece;
- passed pawns or promotion race;
- structural domination;
- transition to a known favorable ending;
- practical error asymmetry.

“I have activity” is incomplete. Which pieces, which squares, for how many tempi, and what can the defender do?

22.3 Three kinds of sacrifice

Combination. Calculation proves a concrete return such as mate or material recovery.

Investment. Compensation is durable but not immediately forced: long-term squares, structure, or initiative.

Speculation. Compensation is uncertain and depends heavily on difficult defense or opponent error.

All three can be practical choices. The proof burden rises from combination to investment to speculation.

22.4 The three-column proof

A sacrifice must survive three broad reply classes:

Opponent choice	Your question	Typical hidden resource
Accept	Is compensation sufficient after the best defense?	return material, trade queens, countercheck
Decline	Did I improve or simply offer material?	keep structure, attack sacrificed piece
Counterattack	Is my threat faster and is my king safe?	zwischenzug, perpetual, attack on queen

Many false sacrifices calculate only acceptance.

22.5 Time decay of compensation

Initiative compensation decays. If you cannot renew threats, the material deficit remains. Structural compensation may persist longer. Passed pawns may grow in value toward the ending. Identify the time profile:

Fast-decay.

Temporary development lead, exposed king before consolidation.

Medium-decay.

Piece activity and space that can be challenged.

Slow-decay.

Permanent pawn weakness, bishop versus fixed color complex, advanced protected passer.

Choose plans that convert fast-decay resources before they vanish.

22.6 Return sacrifice

The defender can often return part of the material to extinguish compensation. This is rational if material was only a means to result utility. Attackers routinely overvalue keeping every sacrificed unit “justified”; defenders overvalue keeping every captured unit.

Worked example: sacrifice to buy a move

You give a pawn to open a file with tempo against the king. The compensation is not “two open lines” in the abstract. It is one immediate rook entry, a forced king move, and a queen arrival before the defender develops. If the defender can return the pawn and trade queens, the initiative may disappear. The sacri-

fice's value is the exact move-order window.

Common error

A beautiful first move does not lower the proof standard. Calculate the opponent's least cooperative response.

Exercise 22.1: Compensation ledger

Analyze a sacrifice using material cost, compensation categories, time profile, acceptance, refusal, counterattack, and possible return sacrifice.

Exercise 22.2: Investment or speculation?

Find two sacrifices: one with durable positional compensation and one relying on a difficult defense. Explain why their discount and risk differ.

The chapter in one minute

A sacrifice trades material now for future resources. Name the compensation, price its decay and uncertainty, and prove the idea against acceptance, refusal, and counterattack.

Endgames as compressed game trees

You will understand why endgames reward backward induction, value king activity and passed pawns dynamically, choose exchanges by terminal state, and use known positions as leaf labels.

The question underneath the chapter

Why can fewer pieces make one tempo and one square more important?

23.1 Fewer pieces, sharper state values

Endgames contain fewer legal interactions, so deeper calculation becomes possible. Yet each tempo, square, and pawn can have larger marginal value. The king changes from protected liability to active piece. Promotion supplies a clear terminal target. Known theoretical positions label leaves as win, draw, or loss.

This makes endgames a natural laboratory for game theory: backward induction, opposition, zugzwang, races, and commitment are visible.

23.2 The endgame evaluation order

Use a different priority than in the middlegame:

1. forcing checks, captures, and promotion threats;
2. king activity and opposition;
3. passed pawns and races;
4. piece activity and restriction;
5. pawn structure and fixed weaknesses;

6. transition to known theoretical positions;
7. practical execution under the clock.

Material still matters, but a distant king or trapped rook can make nominal count misleading.

23.3 The king as a production asset

In the opening, king exposure increases the opponent's forcing options. In the endgame, an active king creates captures, supports passers, and denies squares. The phase transition changes the weight w_K in the evaluation function from safety to activity.

A useful rule is not simply “centralize the king,” but:

Move the king toward the square where it changes the largest number of future pawn and piece outcomes.

23.4 Passed pawns and option value

A passed pawn creates a promotion threat and ties resources. Its value depends on:

- distance to promotion;
- support and blockade;
- whether advancing loses reserve tempi;
- the location of kings and rooks;
- whether it is outside, connected, or protected;
- tactical checks in the promotion race.

Do not automatically push. A pawn can be more valuable as a fixed threat that restricts the opponent than as an advanced target.

23.5 Exchange decisions point to terminal states

“Trade pieces when ahead” is incomplete. Evaluate the state after the final recapture. A piece trade can enter a lost pawn ending. A rook trade can remove your only counterplay. A pawn trade can create an outside passer for the defender.

Use backward comparison:

1. label the resulting ending;
2. locate kings and passers after all recaptures;
3. identify whose move it is;
4. check reserve tempi and pawn races;
5. only then approve the first exchange.

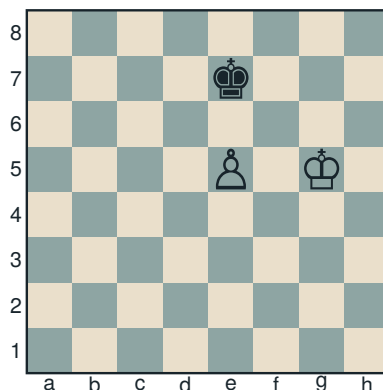


Figure 23.1: In simplified endings, the same placement can change value when the side to move changes. Treat the move as part of the state.

The diagram is intentionally presented as a state-reading exercise rather than a claimed tablebase result. Add the side to move, then calculate candidate king routes and pawn advances. The lesson is that a diagram without turn information is incomplete.

23.6 Known positions as value functions

Endgame study builds a library of exact or reliable leaf values: basic mates, key king-and-pawn positions, rook activity rules, fortress patterns, and standard drawing mechanisms. When calculation reaches one, you can stop expanding and use the stored value.

But retrieve conditions, not slogans. “Rook behind the passed pawn” depends on checks, king placement, and tactical details. A pattern is a theorem with assumptions.

23.7 Practical endgame utility

A theoretically drawn ending can be easy for one side and extremely difficult for the other. A tablebase label does not include clock, memory, or move difficulty. Preserve the objective label, then add execution utility:

- number of only-moves;
- naturalness of defensive setup;
- risk of irreversible pawn moves;
- clock increment;
- familiarity with the technique.

Worked example: simplify into what?

You are up a pawn and can trade queens. The resulting pawn ending leaves your king outside the square of the opponent's passed pawn. The current queen position is risky, but the trade loses by force. "Ahead means trade" fails because the terminal state, not the number of pieces removed, determines utility.

Memory hook

In an endgame, calculate the final recapture and the next king move before celebrating simplification.

Exercise 23.1: Leaf library

List ten endgame positions or techniques you can reliably label as win, draw, or loss. For each, write one condition required for the label.

Exercise 23.2: Exchange backward

Choose an endgame exchange from your game. Reconstruct the position after all recaptures, include side to move, and evaluate king activity and pawn races.

The chapter in one minute

Endgames compress the tree and sharpen tempo value. Activate the king, value passed pawns dynamically, and judge exchanges from the final state backward using known positions as conditional leaf labels.

Match, tournament, and draw strategy

You will translate event scoring into utility, choose risk by standings and color, evaluate draw offers rationally, and separate legitimate event strategy from wishful result chasing.

The question underneath the chapter

What is the value of half a point when half a point changes everything?

24.1 The event changes preferences

In standard individual scoring, a win earns one point, a draw half, and a loss zero, though event rules can differ. The event state adds standings, tie-breaks, remaining rounds, color, and qualification conditions. A move's chess value is filtered through event utility.

Let $x \in \{0, \frac{1}{2}, 1\}$ be the game score and let $G(x, H)$ be event utility given tournament history H . Maximizing x and maximizing G are not always identical. A draw may guarantee qualification; a win may be required for any prize.

24.2 Risk profiles by need

Event need	Position preference	Risk implication
Draw sufficient	robust, low loss probability, simple execution	avoid unnecessary volatility
Win highly valuable	preserve pieces, asymmetry, multiple plans	accept controlled variance
Loss and draw equivalent	maximize win probability, not expected score	sharper gambles may become rational
Unknown standings	maximize ordinary expected score	avoid assumptions
Team context	verify board-order and team incentives	individual utility may differ

Controlled variance means positions where you retain an objective floor and can press, not random self-destruction.

24.3 Draw offers as sequential choices

A draw offer creates an explicit decision with utilities shaped by position, clock, event need, and future risk. Evaluate:

1. objective position and likely result distribution;
2. both clocks and execution difficulty;
3. event value of half a point;
4. whether your emotional state is distorting risk;
5. whether continuing reveals or consumes preparation relevant later.

Do not accept merely because the opponent is higher rated, nor decline merely because you were better earlier. The offer is evaluated from the current state.

24.4 Color and asymmetric utility

Color changes the initial action order, not the value of a win or loss. Yet opening choices and event needs interact with color. With Black in a must-draw game, a sound line with durable equality may dominate an unbalanced line. With White needing a win, avoiding early simplification may increase practical chances, provided objective soundness remains.

24.5 Match games and dynamic strategy

A match score creates state-dependent utilities. Leading by one point late in a match increases the value of low-variance choices; trailing increases the value of preserving winning chances. Preparation also repeats, so repertoire mixing and information management matter.

A useful match table:

Score state	Win utility	Draw utility	Loss cost
Leading late	high	very high	very high
Tied	high	medium	high
Trailing late	very high	low	medium

The words are intentionally qualitative; exact utility depends on remaining games and advancement rules.

24.6 Ethics and rule compliance

Strategic event thinking must remain within fair play and event rules. Do not assume cooperation from other players, manipulate results through prohibited arrangements, or use outside assistance. Game theory describes incentives; it does not excuse misconduct.

Worked example: the forced draw you dislike

You had a winning position, blundered, and can now force perpetual check. Declining the draw line offers only a small chance of winning and a large chance of losing. If half a point meets your event need, the perpetual has high utility. The earlier win is a sunk state. Rational choice can feel disappointing and still be correct.

Common error

“I need to win” changes the ranking of sound options. It does not make a refuted move sound.

Exercise 24.1: Event utility

Create qualitative win/draw/loss utilities for four scenarios: first round, final round needing a draw, final round needing a win, and two games left while leading a match.

Exercise 24.2: Draw-offer decision

Analyze a real or hypothetical draw offer using position, clocks, event value, emotional state, and preparation cost. State the decision and its assumptions.

The chapter in one minute

Event state changes utility. Translate standings and match score into risk preferences, evaluate draw offers from the current state, and let result need choose among sound options rather than justify unsound ones.

Part VI

Practice Until It Is Automatic

The post-game learning loop

You will review decisions before consulting an engine, separate process errors from knowledge errors, update your model from counterfactuals, and convert one game into a small set of reusable rules.

The question underneath the chapter

How do you learn from a game without merely collecting the engine's preferred moves?

25.1 A result is a noisy label

Winning does not prove your decisions were good; losing does not prove they were bad. The result is one terminal sample from a tree containing missed opportunities, robust choices, and later errors. Review should estimate decision quality, not reward hindsight.

The unit of learning is the **critical decision**: a state where candidates differed materially in utility, where you spent significant time, or where evaluation changed.

25.2 The five-pass review

Pass 1: reconstruct without assistance. Record what you thought, what candidates you considered, your clock, and your evaluation. Do not rewrite memory to match the truth.

Pass 2: opponent model. For each critical state, identify the opponent's best plan and best response. Mark surprises and what they revealed.

Pass 3: counterfactual tree. Analyze the candidates you rejected. Use a small tree and stable leaves. State what evidence would have changed your choice.

Pass 4: external verification. Consult an engine, database, coach, or endgame reference. Use the tool to test hypotheses, not to replace them. Ask why the recommendation works and what refutes your move.

Pass 5: compress. Create one pattern card, one process guardrail, and one knowledge task. More than three takeaways often means none will be retained.

25.3 Error taxonomy

Classify the failure before prescribing study.

Error type	Diagnostic	Training response
State error	missed piece, right, threat, or side effect	board scan and visualization
Candidate error	best move never entered the set	CCT+I and opponent free-move drills
Reply error	assumed cooperative response	best-response and SAFE drills
Calculation error	tree branch or move order wrong	short forcing-line calculation
Evaluation error	leaf features weighted poorly	comparative position study
Utility error	event, clock, or risk mis-ranked	pre-game utility card
Time error	thinking budget misallocated	clock map and stop rules
Belief error	over- or under-updated opponent model	evidence ledger
Execution error	knew idea but failed technique	endgame or conversion drills

A tactical puzzle will not cure a utility error. Opening memorization will not cure a reply-first failure.

25.4 Counterfactual discipline

Ask three counterfactual questions:

1. If the opponent had chosen their best reply, would my move still be acceptable?
2. If my evaluation of one feature were wrong, would the move remain robust?

3. If I had half the time, which simpler process would have reached the decision?

Counterfactuals reveal causality. Perhaps the move won only because the opponent blundered; perhaps it was robust despite losing later.

25.5 Using engines without outsourcing thought

An engine is strongest as a falsifier and comparator.

- Compare your top three candidates, not only the played move.
- Hide the principal variation at first; guess the refutation.
- Ask which evaluation component changed.
- Check at several depths for tactical stability.
- Translate the line into a verbal reason and a trigger condition.
- Revisit the position days later without the engine.

Do not memorize decimal evaluations. A shift from $+0.4$ to -0.8 matters because of the underlying tactical or strategic reason, not because the digits are magical.

Worked example: the wrong lesson from a blunder

You lose after missing a knight fork. Engine review shows your previous pawn break was also inferior. The tempting takeaway is “study knight forks.” Error taxonomy shows two failures: a reply scan missed the fork, and an evaluation error overvalued space from the pawn break. Create two tasks: a SAFE trigger for loose king-and-queen geometry, and a structural study of the break. One label would hide half the lesson.

25.6 Pattern cards and spaced retrieval

A useful card has four lines:

1. **Trigger:** the visible configuration.
2. **Question:** what to ask.
3. **Action:** typical candidate or method.
4. **Exception:** condition that makes it fail.

Example: Trigger—queen and king share a knight-fork geometry. Question—can a tempo move create the fork? Action—scan checks and attacks on queen. Exception—the knight is pinned or the destination square is defended tactically.

Review cards after one day, one week, and one month. Retrieval is the practice; rereading is not enough.

25.7 Process metrics

Track controllable behaviors rather than rating alone:

- percentage of critical moves where you named an opponent best reply;
- number of blunders caught by SAFE before moving;
- time spent on truly critical decisions;
- number of games reviewed before engine use;
- recurrence rate of the same error category;
- conversion and defense outcomes by position type.

These metrics are diagnostic, not moral scores.

Memory hook

A game is learned when it changes a future question, not when its engine line has been copied.

Exercise 25.1: Five-pass review

Apply the five passes to one complete game. Produce exactly three outputs: a pattern card, a process guardrail, and a knowledge task.

Exercise 25.2: Error taxonomy

Classify the five most important mistakes in a game. Which category recurred, and what training intervention directly targets it?

The chapter in one minute

Review decisions, not only results. Reconstruct first, build counterfactuals, verify last, classify the error, and compress the game into one pattern, one guardrail, and one knowledge task.

A twelve-week course in thinking

You will turn the book into a sustainable study program, practice one cognitive skill at a time, measure transfer to games, and integrate the full decision loop.

The question underneath the chapter

How do ideas become board habits?

26.1 The training principle

Knowledge becomes skill through retrieval under conditions resembling play. Each week therefore contains four elements:

1. **Read:** understand one lens.
2. **Drill:** practice it in isolated positions.
3. **Play:** use one process goal in real games.
4. **Review:** measure whether the process appeared.

Do not try to “use the entire book” on every move. Automation comes from narrow focus followed by integration.

26.2 Weekly plan

Wk	Focus	Drill	Game process goal
1	Frame and state	reconstruct positions from notation; list missing facts	say side to move, material, checks, loose pieces
2	Utility and evaluation	rate M, K, A, P, S, T, O in ten positions	name result need and dominant feature

3	Candidates	CCT+I sets without moving pieces	generate at least three distinct functions
4	Best responses	opponent free-move and SAFE drills	state strongest reply before moving
5	Minimax	compare security levels in short trees	reject tactical cliffs
6	Backward induction	mate, promotion, and material terminal targets	stop at stable leaves
7	Bounded search	timed candidate triage	spend time by volatility and irreversibility
8	Expected utility	rank equal positions by error asymmetry	choose practical robustness among sound moves
9	Beliefs and prophylaxis	evidence ledgers and four responses to plans	ask what the last move revealed and enabled
10	Openings and mid-dlegames	convert lines to trees; imbalance maps	leave memory when conditions change
11	Endgames and event utility	final-recapture and result-need exercises	calculate terminal transition before exchange
12	Review and integration	full five-pass review	use MODEL-REPLY-VALUE-DECIDE-UPDATE

26.3 A 45-minute study session

- 1. **5 minutes—retrieval.** Without notes, write last session’s memory hook and process rule.
- 2. **10 minutes—concept.** Read one subsection and restate it in your own words.
- 3. **15 minutes—deliberate drill.** Solve three to five positions with written candidates and replies.
- 4. **10 minutes—one game fragment.** Apply the skill to your own position.
- 5. **5 minutes—compression.** Create one card and schedule its review.

Longer sessions can add play or a full game review. Preserve the order: retrieve before rereading and think before engine verification.

26.4 Skill ladders

Progress by increasing one difficulty dimension at a time.

Candidate ladder. One forcing position → quiet position → mixed tactical-strategic position → low-time selection.

Calculation ladder. One reply → two branches → intermediate move → quiet leaf evaluation → clock constraint.

Opponent-model ladder. Identify plan → identify best response → infer belief → update from new evidence → adapt in a rematch.

26.5 The integrated move protocol

By week twelve, use **BOARD**:

- B Board facts.** What changed? Checks, captures, threats, loose pieces, evaluation vector.
- O Opponent.** What do they want, what is their best reply, and what did the last move reveal?
- A Actions.** Generate three to five candidates with distinct functions.
- R Replies and results.** Search forcing branches, stable leaves, security levels, and practical error asymmetry.
- D Decide and update.** Choose by current utility, run SAFE, move, and rebuild the state.

BOARD is the same book in one word. Under time pressure, compress it to:

Facts → their threat → two candidates → SAFE → move.

26.6 Assessment positions

At the end of weeks 4, 8, and 12, select ten positions from your own games. For each, record:

- state evaluation without an engine;
- candidate set;
- strongest reply to each serious candidate;
- chosen move and confidence;

- time used;
- later verification;
- error category, if any.

Compare process accuracy, not only move match. You are improving when candidate sets are richer, replies are more hostile, time is better allocated, and repeated error categories decline.

26.7 Retention schedule

Every important concept should be retrieved at expanding intervals:

same day → 1 day → 1 week → 1 month → before an event.

Mix old and new positions. A skill remembered only in the chapter where it was learned is context-bound; chess requires transfer.

Worked example: one-focus game

This week's goal is best responses. You play a rapid game and, before every non-forced move, name one strongest reply. You still make strategic mistakes, but you catch two tactical blunders. In review, measure whether the behavior occurred, not whether every move was best. The habit is being installed; other layers return in later weeks.

26.8 Graduation standard

You have internalized the framework when, in a difficult position, you naturally:

- reconstruct before reacting;
- generate before calculating;
- search the opponent's best reply;
- compare stable leaves with the right utility;
- spend time where value can flip;
- update without defending your previous story.

No protocol removes mistakes. It makes errors visible, classifiable, and trainable.

Exercise 26.1: Build your course

Put actual dates, game days, and review sessions onto the twelve-week plan. Choose one process metric for each four-week block.

Exercise 26.2: Final integration

Analyze one critical position using the full BOARD protocol. Then repeat under a two-minute limit and compare what was preserved or lost.

The chapter in one minute

Train one lens at a time through retrieval, drills, play, and review. Integrate with BOARD: Board facts, Opponent, Actions, Replies/results, Decide/update.

Part VII

Reference and Solutions

Notation and rules reference

A.1 Board coordinates

Files are labeled a through h from White’s left to right. Ranks are labeled 1 through 8 from White’s side toward Black’s. A square combines file and rank, such as e4.

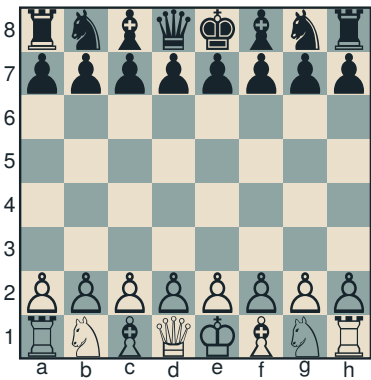


Figure A.1: Coordinates and the initial position, viewed from White’s side.

A.2 Piece symbols and names

White	Name	Black	Letter
♔	King	♚	K
♕	Queen	♛	Q
♖	Rook	♜	R
♗	Bishop	♝	B
♘	Knight	♞	N
♙	Pawn	♟	no letter

Knights use N because K is reserved for the king. Pawn moves use only the destination square.

A.3 Algebraic notation

Notation	Read as	Meaning
e4	e-four	pawn moves to e4
Nf3	knight f-three	knight moves to f3
Rae1	a-rook e-one	rook from the a-file moves to e1
Qxh7+	queen takes h-seven check	queen captures on h7 and checks
Re8#	rook e-eight mate	rook moves to e8 and gives check-mate
0-0	castles kingside	king and rook castle on the king-side
0-0-0	castles queenside	king and rook castle on the queen-side
e8=Q	e-eight promotes to queen	pawn reaches e8 and promotes
exd6 e.p.	e-pawn takes d-six en passant	special capture after a two-square pawn move
1-0	White wins	final result
0-1	Black wins	final result
$\frac{1}{2} - \frac{1}{2}$	draw	final result

A plus sign marks check; a hash marks checkmate. An “x” marks capture. Annotation symbols such as !, ?, !?, and ?! express judgment and are not part of legal notation.

A.4 Essential rule reminders

Check and mate.

A player in check must remove it. Checkmate ends the game.

Stalemate.

A player with no legal move who is not in check is stalemated; the game is drawn.

Castling.

Castling requires unmoved king and rook, empty intervening squares, no current check, and no king passage through an attacked square.

En passant.

A pawn that advances two squares and lands beside an opposing pawn may be captured as though it moved one square, but only on the immediately following move.

Promotion.

A pawn reaching the final rank promotes to queen, rook, bishop, or knight.

Repetition and move-count draws.

Formal claim and automatic-draw details depend on the governing laws and platform. Record history when relevant and consult the event rules.

Touch move and clock procedure.

Over-the-board events may enforce touch-move and specific clock rules. Follow the organizer’s regulations.

A.5 Nominal material values

Piece	Common prior	Main strength	Main dependence
Pawn	1	structure, promotion	advancement and support
Knight	3	forks, closed positions	stable central squares
Bishop	3	long range, both wings	open diagonals
Rook	5	files, ranks, passers	entry squares and activity
Queen	9	coordinated multi-threats	king safety and tempo
King	terminal	survival, endgame activity	phase of game

These are priors, not exchange commands. Functional value depends on the position.

A.6 A compact analysis notation

For private notes, use:

- $\rightarrow$ for “leads to”;
- $\pm$ for White clearly better, $\mp$ for Black clearly better;
- ∞ for unclear compensation;
- M, K, A, P, S, T, O for material, king safety, activity, pawn structure, space, time, options;
- F, N, D for forced, natural, and difficult-to-find replies;
- W, D, L for win, draw, and loss outcome classes.

Decision sheets and printable templates

B.1 Pre-game utility card

Event and preference frame

Time control: _____ Color: _____

Event state and result need: _____

Position types I understand well: _____

Clock reserve rule: _____

Emotional guardrail: _____

B.2 BOARD position sheet

B — Board facts

Material: _____

Checks, captures, threats, loose pieces: _____

M, K, A, P, S, T, O: _____

What changed on the last move? _____

O — Opponent

Their free move / plan:_____

Strongest reply to my likely move:_____

Evidence about preparation, style, or risk:_____

A/R/D — Actions, replies, decision

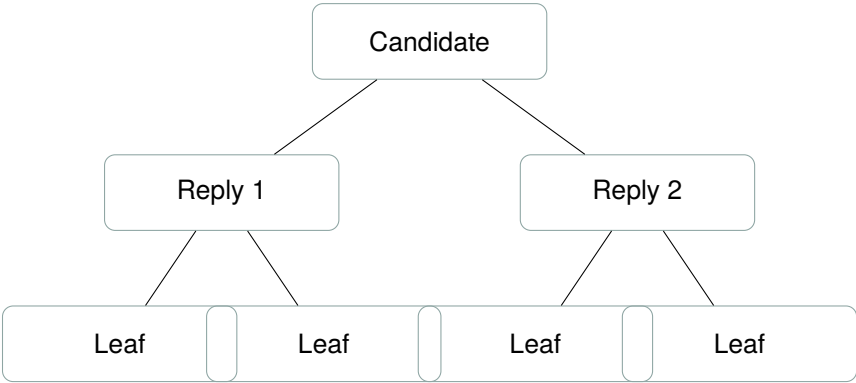
	Candidate 1	Candidate 2	Candidate 3
Function			
Best			
reply			
Stable			
leaf			
Risk /			
ease			
Chosen move and why:			
SAFE result:			

B.3 Calculation tree sheet

Terminal claim

What final state would prove the line?_____

What counts as a stable leaf?_____



Comparison

Worst credible leaf:_____

Most likely leaf:_____

Unresolved forcing feature:_____

Confidence and stop rule:_____

B.4 Post-game five-pass sheet

- Pass 1: Reconstruct.** My candidates, evaluation, clock, and thought process.
- Pass 2: Opponent.** Their best response, plan, and evidence revealed.
- Pass 3: Counterfactual.** Rejected candidates and stable leaves.
- Pass 4: Verify.** Engine, database, coach, or reference; reason for difference.
- Pass 5: Compress.** One pattern card, one guardrail, one knowledge task.

Error classification

State / Candidate / Reply / Calculation / Evaluation / Utility / Time / Belief / Execution

Dominant category:_____

Direct training intervention:_____

B.5 Pattern card

Reusable pattern

Trigger configuration:_____

Question to retrieve:_____

Typical action or method:_____

Exception / refutation condition:_____

Review dates: same day 1 day 1 week 1 month

Solutions and answer guidance

Many exercises use your own games, so a good answer is defined by method rather than one move. The solutions below give the required reasoning structure and a representative response.

Chapter 1

Exercise 1.1. Perfect information concerns the observable game state: piece locations, legal rights, history, and clocks. Beliefs concern hidden human variables such as whether the opponent knows a line or has seen a tactic. Example board fact: Black's queen is undefended. Hidden human fact: Black may or may not have noticed it.

Exercise 1.2. A complete answer names the failed stage. Example: MODEL omitted an overloaded defender; REPLY assumed a passive recapture; VALUE stopped before a countercheck; DECIDE favored excitement over result; UPDATE continued to use the old evaluation after the exchange.

Chapter 2

Exercise 2.1. Frame: White has 42 seconds; castling rights are lost. Preference: White dislikes queenless endings; Black enjoys complications. "A draw wins the match" enters the event frame and thereby changes the ranking of terminal results; it should be recorded before utility is assigned.

Exercise 2.2. At minimum: side to move, castling/en-passant rights, repetition or move-count relevance, clock times and time control, event result need, and any missing prior move needed to establish legal rights.

Chapter 3

Exercise 3.1. A contingent strategy should specify how the isolated pawn will be fixed and attacked, plus responses to advance, defense, exchange, or counterplay. Example: blockade if it advances, double rooks if defended passively, switch wings if pieces become tied, and abandon the target if a tactical attack on your king appears.

Exercise 3.2. A valid set includes legal examples from all requested functions. The learning objective is breadth before commitment. After generation, run reply-first triage; do not reject quiet improvements merely because a check exists.

Chapter 4

Exercise 4.1. In training, a difficult but instructive ending can outrank an easy draw. In a must-draw game, low loss probability and simple execution rank highly. In a must-win game, a sharp but sound winning chance can outrank a sterile small edge. The rankings must be internally consistent and tied to event utility.

Exercise 4.2. The answer should identify the emotional preference, the chess utility it displaced, and a prewritten guardrail. Example: pride rejected perpetual check; guardrail: “After losing an advantage, rank only outcomes reachable from the current state.”

Chapter 5

Exercise 5.1. Rate all seven coordinates, then generate actions from weaknesses. If activity is unfavorable, improve or exchange the worst piece. If king safety is unfavorable, neutralize checks, add defenders, close lines, or trade queens. A complete answer connects every negative feature to at least one candidate.

Exercise 5.2. Compare mobility, targets, defensive duties, pawn colors, open lines, and the position after recapture. Nominal equality does not settle the exchange. The answer should state which future function was gained or lost.

Chapter 6

Exercise 6.1. The free move should express the opponent’s strongest plan. Two responses may stop it directly, permit it while gaining elsewhere, provoke it as a weakness, or race it. The key is to compare response cost with plan value.

Exercise 6.2. SAFE must inspect checks, captures, threats, and the end of the forcing sequence. A strong answer names the exact category and the changed relationship, such as a queen move abandoning back-rank defense.

Chapter 7

Exercise 7.1. Create reply columns such as forcing defense, simplification, and counterattack. One move dominates only if it is never worse across the relevant columns and better in at least one. Otherwise state the condition on which the ranking depends.

Exercise 7.2. Typical examples include quiet retreats, underpromotions, clearance moves, and intermediate moves. Name the misleading principle and the hidden resource. The lesson is to store principles with exceptions.

Chapter 8

Exercise 8.1. For each candidate, take the minimum value among the opponent's strongest replies. Choose the candidate with the larger minimum. If rough values are speculative, state the uncertainty rather than manufacture precision.

Exercise 8.2. Reply robustness asks whether one defense collapses the move; evaluation robustness asks whether a mistaken feature estimate changes the choice; execution robustness asks whether only-moves follow. The weakest dimension indicates where more calculation or a safer move is needed.

Chapter 9

Exercise 9.1. State a terminal result such as mate, decisive material, promotion, forced draw, or a known ending. Enumerate king moves, captures, blocks, counterchecks, refusal, and counter-threats as relevant. Every branch must reach a leaf supporting the claim.

Exercise 9.2. A threat succeeds only if the opponent passes or responds inadequately. A forced line survives every relevant defense. The answer should name the defender's available choice in the threat example and the lack of a utility-preserving choice in the forced example.

Chapter 10

Exercise 10.1. For each preparation move, list its benefit and the opponent's best use of the tempo. Preparation is justified only if the net dynamic gain is positive and no forcing counterplay appears.

Exercise 10.2. Conversion examples: win material, damage structure, force a favorable ending. Renewal examples: create another forcing threat or bring a piece with tempo. Defense examples: trade attackers, return material, close lines, or countercheck.

Chapter 11

Exercise 11.1. Allocate more time to candidates high in forcingness, volatility, irreversibility, or uncertainty. A valid two-minute split might be 70 seconds on a king-opening pawn break, 35 seconds on a tactical exchange, and 15 seconds on a reversible improvement, leaving 20 seconds for comparison and SAFE.

Exercise 11.2. Extend until immediate checks, captures, promotions, and trapped-piece issues are resolved. Name the horizon cause, such as temporary material gain or a forcing reply just beyond the original leaf.

Chapter 12

Exercise 12.1. Probabilities must sum to one. Compute $EU = \sum p_i u_i$, then vary the most uncertain probability. If the preferred move changes after a small adjustment, the decision is fragile and objective robustness deserves more weight.

Exercise 12.2. Count acceptable moves and classify them as natural or difficult. The side with several coherent improvements and fewer tactical cliffs has favorable error asymmetry, even if the objective evaluation is equal.

Chapter 13

Exercise 13.1. Examples: always playing one defense, refusing queen trades, moving instantly when prepared, or accepting every draw offer. For each, describe the exploitation. Keep a habit when its learning value or objective quality exceeds predictability cost; otherwise broaden or improve it.

Exercise 13.2. Both options must be objectively sound, understood, and affordable to study. Give them distinct roles, such as a forcing main line and a quieter structural alternative, plus intended frequencies across repeated games.

Chapter 14

Exercise 14.1. Compare actual time with decision importance. Overspending on forced recaptures and underspending on irreversible pawn breaks are classic errors. The answer should propose a reallocation and preserve a future reserve.

Exercise 14.2. A strong five-step routine checks both kings, loose pieces, one forcing and one robust candidate, SAFE, and physical clock execution. The reserve rule should be specific to the time control.

Chapter 15

Exercise 15.1. State a hypothesis and an alternative. The update should depend on likelihood ratio: an instant accurate novelty response is more diagnostic of preparation than an instant forced recapture. Preserve uncertainty if both explanations predict the evidence similarly.

Exercise 15.2. Beliefs can choose between close sound candidates, such as theory versus a fresh structure. They cannot turn a forced loss into a good move. State the objective floor first and the belief adjustment second.

Chapter 16

Exercise 16.1. Separate square control, lines, weaknesses, and threats from intimidation or perceived intention. Only board effects remain against a fearless opponent. A strategically valid move must retain sufficient value there.

Exercise 16.2. A credible threat is beneficial if ignored, cannot be answered while improving freely, remains executable after the reply, does not assume cooperation, and costs less than it gains. Classify the actual threat accordingly.

Chapter 17

Exercise 17.1. Stopping removes the plan; permitting accepts it as harmless; provoking encourages it because it concedes something; racing ignores it for faster counterplay. Compare resulting utilities, not the emotional comfort of prevention.

Exercise 17.2. A complete second-order answer states your plan, the opponent's belief about it, the preventive action that belief induced, and a counter-plan that exploits the prevention.

Chapter 18

Exercise 18.1. Classify the main process distortion, then choose a direct guardrail. Confirmation bias: search for one refutation. Anchoring: rebuild the vector. Sunk cost: evaluate only current futures. Action bias: include a waiting move.

Exercise 18.2. Each story becomes observable claims. "Attack" becomes attackers, defenders, entry squares, and forcing moves. "Weak pawn" becomes fixed, attackable, unable to advance, and costly to defend.

Chapter 19

Exercise 19.1. List mutual learning and predicted adaptation. Example: they learned your narrow defense; you learned they dislike closed centers; they may prepare the forcing main line; you may diversify into a sound closed structure.

Exercise 19.2. Stable traits require repetition across positions; temporary states include fatigue and time trouble; position effects follow directly from board necessity. Give confidence and avoid using one game as decisive evidence.

Chapter 20

Exercise 20.1. The tree should include at least three opponent choices, your purpose-based response, and an invalidating condition. This converts sequence memory into contingent knowledge.

Exercise 20.2. A good portfolio usually has a stable core for pattern depth and one sound alternative for adaptability. The answer should acknowledge study cost and not add variety beyond competence.

Chapter 21

Exercise 21.1. List asymmetries, choose the dominant one, and design two plans with explicit target, worst-piece improvement, counterplay, commitments, and transition. Rank plans by net portfolio rather than activity alone.

Exercise 21.2. Every piece receives an attack, restrict, defend, transform, or reserve role. The worst piece is the one without a useful current or future job; propose a legal route and test the opponent's tempo.

Chapter 22

Exercise 22.1. The ledger must include immediate material, exact compensation, time decay, all three reply classes, and a possible return sacrifice. The conclusion should state whether compensation is forced, durable, or mainly practical.

Exercise 22.2. An investment has identifiable long-lived resources such as permanent weakness or protected passers. A speculation depends on difficult defense or opponent error, so delayed benefits receive a larger discount and risk penalty.

Chapter 23

Exercise 23.1. Examples include basic mates, key pawn positions, opposition, rook cut-off methods, fortress patterns, and standard promotion races. Each label needs a condition such as side to move, king placement, checking distance, or absence of a tactical exception.

Exercise 23.2. Reconstruct after all captures, then include side to move, king routes, passers, reserve tempi, and races. The initial exchange is good only if the terminal state has acceptable utility.

Chapter 24

Exercise 24.1. First round normally uses ordinary expected score. Must-draw heavily values low loss probability. Must-win emphasizes win probability among sound choices. A late match lead raises draw utility and loss cost. Exact rankings depend on event rules.

Exercise 24.2. Evaluate current position, result distribution, clocks, event value, emotion, and information cost. State assumptions. Earlier advantage and rating are not sufficient reasons to accept or reject.

Chapter 25

Exercise 25.1. The final output is deliberately limited: one trigger-question-action-exception card, one behavioral guardrail, and one specific knowledge task. Verification should follow unaided reconstruction and counterfactual analysis.

Exercise 25.2. Classify five decisions, count recurring categories, and prescribe the matching drill. A reply error calls for best-response work; a time error for budgeting; an evaluation error for comparative positions.

Chapter 26

Exercise 26.1. Assign dates and game opportunities to all twelve weeks. Suitable four-week metrics include reply naming rate, SAFE catches, time-allocation accuracy, or percentage of games reviewed before engine use.

Exercise 26.2. The full version should include all BOARD stages. Under two minutes, preserve board facts, opponent threat, two candidates, strongest replies, and SAFE. Note which deeper evaluation or belief work was lost and whether the compressed decision remained robust.

Glossary of chess decision terms

Action.

One legal move selected at the current state.

Action set.

All legal moves available in a state; the effective set excludes moves made unusable by tactical cost.

Anchoring.

Remaining attached to an earlier evaluation despite new evidence.

Backward induction.

Evaluating terminal or stable states first and carrying their values backward through the tree.

Bayesian updating.

Revising beliefs when new evidence is more likely under one hypothesis than another.

Belief.

A probability or confidence assigned to a hidden opponent characteristic or future response.

Best response.

The reply that maximizes a player's utility against a specified opposing choice.

BOARD.

Board facts, Opponent, Actions, Replies/results, Decide/update.

Bounded rationality.

Rational decision making under limited time, memory, and computation.

Candidate move.

A legal action selected for serious comparison.

CCT+I.

Checks, captures, threats, and improvements; a candidate-generation order.

Cheap talk.

Communication with little direct board cost or commitment.

Commitment.

An action that removes future options to coordinate or make a plan credible.

Common knowledge.

A fact known by both players, with each aware that the other knows it.

Compensation.

Nonmaterial resources received in exchange for material, such as time, activity, structure, or king exposure.

Constant-sum.

A result structure in which one player's gain corresponds to the other's loss.

Counterfactual.

Analysis of what would have happened after an unchosen action or different reply.

Credible threat.

A threatened future action that remains beneficial to execute if ignored.

Decision state.

The complete practical situation from which a move is chosen.

Deterministic.

A property of a game in which actions have fixed transitions rather than random outcomes.

Dominance.

A relation in which one action is never worse and sometimes or always better across specified replies.

Dynamic value.

Value arising from move order, initiative, threats, and timing rather than static features alone.

Effective action set.

Legal moves that remain strategically viable after obvious tactical costs are considered.

Equilibrium.

Strategies that are mutual best responses; neither player benefits by changing alone.

Error asymmetry.

A difference in how many natural or precise moves each side must find.

Evaluation vector.

A multidimensional summary such as material, king safety, activity, pawn structure, space, time, and options.

Expected utility.

Probability-weighted value over possible outcomes.

Exploitability.

The gain an informed opponent can obtain from knowing your predictable choice rule.

Extensive form.

A representation of sequential decisions as a branching tree.

Forcing move.

A move that sharply restricts the opponent's acceptable replies.

Frame.

Players, states, actions, transitions, rules, and terminal outcomes before preferences are added.

Functional value.

A piece or pawn's value from the jobs it performs in the current state.

Game tree.

The branching structure of moves, replies, and subsequent positions.

Guardrail.

A precommitted process rule designed to interrupt a known bias or error.

Horizon.

The point where calculation stops and evaluation begins.

Horizon effect.

A mistake caused by stopping just before a decisive consequence.

Imbalance.

A persistent asymmetry that can support a plan.

Incomplete information.

Uncertainty about hidden types, intentions, knowledge, or other non-board variables.

Information set.

Decision nodes a player cannot distinguish; in formal chess, visible board states are distinguishable.

Initiative.

Temporary control over the opponent's priorities through threats and forcing questions.

Intertemporal exchange.

Trading a resource now for compensation arriving later.

Iterated elimination.

Repeatedly removing dominated or refuted choices as analysis progresses.

Leaf.

A terminal or chosen stopping point in a calculation tree.

Mechanism design.

Structuring available choices so another player's best response advances your objective.

Minimax.

Choosing the action with the best worst-case value under optimal resistance.

Mixed strategy.

A probability distribution over multiple actions or plans.

Negamax.

A symmetric minimax formulation using value from the player-to-move's perspective.

Nominal value.

A general piece-value prior before state-specific functions are considered.

Objective floor.

The value of a candidate against the strongest reply found.

Opening prior.

Stored theory and experience that guide initial attention among moves.

Option value.

Value created by preserving useful future choices.

Outcome.

A terminal result or evaluated future position.

Perfect information.

A property in which the formal game state and action history relevant to play are observable.

Posterior.

A belief after new evidence has been incorporated.

Practical ceiling.

A candidate's upside against likely human responses, after its objective floor is acceptable.

Practical utility.

Value adjusted for clock, complexity, familiarity, and human error.

Preference.

A ranking over outcomes.

Prior.

A belief held before the current evidence is observed.

Prophylaxis.

Reasoning about the opponent's intended action and choosing whether to stop, permit, provoke, or race it.

Pure strategy.

A definite action rule at every state.

Quiescent position.

A leaf where immediate forcing actions have settled enough for static evaluation.

Repeated game.

A sequence of encounters in which past behavior changes future beliefs and incentives.

Reserve tempo.

A move saved to transfer the obligation to act at a critical later moment.

Robust move.

A move that remains acceptable across replies, evaluation uncertainty, or execution errors.

SAFE.

Scan checks, Assess captures, Find threats, Evaluate the end.

Security level.

The worst outcome a strategy can guarantee against the replies considered.

Selective deepening.

Searching important branches more deeply than quiet or settled branches.

Signal.

An observed action that may convey information about intention or type.

Stable leaf.

A calculation endpoint with resolved forcing actions and a reliable evaluation.

State.

All facts relevant to legal actions, transition, and practical choice.

Strategy.

A contingent rule specifying action at every state that may be reached.

Tempo.

One unit of move order whose value depends on the state.

Terminal state.

A checkmate, draw, resignation result, or other game-ending state.

Threat.

A prospective action that gains value if the opponent does not respond adequately.

Transposition.

Different move orders reaching the same position.

Type.

A hidden characteristic such as preparation, style, or risk tolerance.

Utility.

A numerical representation of preference ordering.

Volatility.

The degree to which one move can sharply change evaluation or result.

Zugzwang.

A state in which every legal move worsens the player's position, so the obligation to act is costly.

Conclusion: See the reply

Chess thought becomes clearer when you stop treating a move as an object and start treating it as an intervention in a decision system.

The board supplies a state. Preferences tell you what outcomes matter. Candidate moves alter the opponent's choices. Best responses test your ideas. Minimax protects against cliffs. Backward induction proves forcing lines. Expected utility distinguishes theoretically equal practical choices. Beliefs adapt to people without replacing board truth. Repeated play turns habits into information. Review updates the model you will carry into the next game.

The entire framework can be retained as five imperatives:

- 1. Model before moving.**
- 2. Give the opponent agency.**
- 3. Compare branches at stable leaves.**
- 4. Choose with the right utility and clock.**
- 5. Update instead of defending the old story.**

A strong player is not someone who never miscalculates. A strong player notices which uncertainties matter, searches where value can flip, and learns in a way that changes future decisions.

The final memory hook

The move is not the point. The point is the tree it creates, the reply it survives, and the future choices it preserves.

One-page field guide

BOARD

B — Board facts: What changed? Material, checks, captures, threats, loose pieces, *M*, *K*, *A*, *P*, *S*, *T*, *O*.

O — Opponent: What is their free move, strongest reply, and likely plan? What did the last move reveal?

A — Actions: Generate checks, captures, threats, and improvements. Include distinct functions.

R — Replies/results: Search the least cooperative reply, continue to a stable leaf, compare security level and error asymmetry.

D — Decide/update: Use current event utility and clock, run SAFE, move, then rebuild the state.

When the clock is low

Both kings' checks → loose pieces → their threat → one forcing and one robust candidate → SAFE → move.

When reviewing

Reconstruct unaided → model the opponent → build counterfactuals → verify externally → create one pattern card, one guardrail, and one knowledge task.

Method and source note

This textbook is an original educational synthesis written for the reader's request. Its conceptual sequence was inspired by the reader-supplied game-theory textbook *GT_book.pdf*: begin with a game frame, add preferences and utility, compare strategic actions, reason through extensive-form trees, then incorporate uncertainty, beliefs, and incomplete information. The chess explanations, frameworks, acronyms, diagrams, examples, exercises, and solutions here are newly written and do not reproduce the source text.

Game-theoretic notation is used as a thinking aid. Chess evaluations, probabilities, and piece values are intentionally described as context-dependent heuristics unless a terminal result is proved. For exact event procedures and draw claims, consult the official rules governing the competition.